Article

Adversarial Prompt Sensitivity in CLIP-Family Models for Zero-Shot Classroom Behavior Analysis

Yan Ma [1,2], Lizhuo Zhang [3*], and Xinjie Wu [4]

[1] School of Foreign Studies, Changsha University of Science and Technology, Changsha 410114, China; mayan@csust.edu.cn
[2] School of Education, Hunan Agricultural University, Changsha 410128, China
[3] School of Information and Intelligence, Hunan Agricultural University, Changsha 410128, China; zhanglizhuo@hunau.edu.cn
[4] College of Liberal Arts and Social Sciences, City University of Hong Kong, Hong Kong, China
* Correspondence: zhanglizhuo@hunau.edu.cn

Abstract

Vision–language foundation models such as CLIP are increasingly used for zero-shot behavior recognition, yet their robustness to prompt variations remains poorly understood. This paper investigates prompt sensitivity as an adversarial vulnerability in CLIP-family models for zero-shot classroom behavior analysis. Five representative vision–language models (CLIP/OpenAI, OpenCLIP/LAION, SigLIP2, EVA02-CLIP, and DFN-CLIP) are evaluated on three public classroom behavior benchmarks under a strict symmetric protocol. We compare four generic prompt strategies with a training-free Class-Aware Prompt Ensemble (CAPE), treating prompt wording as a controlled perturbation. Results show that minor prompt changes can cause catastrophic performance degradation. On SigLIP2, an alternative wording of CAPE reduces Hit@1 on TeacherBehavior from 95.5% to 31.4%, a 64.1 percentage-point drop that exceeds the differences between model backbones. Across all five models, action-oriented prompts improve Hit@1 by up to 54 percentage points compared with label-only prompts. We further demonstrate that the apparent superiority of zero-shot CLIP over supervised linear probes largely arises from metric asymmetry. While zero-shot methods achieve higher Hit@1, they consistently underperform linear probes in multi-label evaluation (Sample-F1: 60–66% vs. 88–90%; Macro-F1: 49–61% vs. 68–78%). Bootstrap confidence intervals and paired-bootstrap significance tests further show that several reported performance differences are not statistically significant. These findings reveal prompt sensitivity as a fundamental deployment risk for vision–language foundation models in domain-specific behavior analysis. Prompt variations involving only a few words can silently undermine recognition performance while remaining hidden by conventional evaluation metrics. We therefore recommend that future benchmark studies report prompt configurations, multi-label F1 scores, and uncertainty estimates alongside headline Hit@1 to surface rather than conceal this vulnerability.

Keywords: prompt sensitivity; adversarial robustness; foundation models; zero-shot classification; CLIP; behavior analysis; vision–language models; prompt engineering; classroom activity recognition

Received:
Revised:
Accepted:
Published:

1. Introduction

Automatic recognition of classroom teaching activities is essential for educational quality assessment, teacher professional development, and intelligent classroom analytics [1,2].

 https://doi.org/10.3390/sym1010000

In modern smart classrooms, camera systems continuously capture teaching and learning behaviors for downstream analysis [3]. Developing robust image-based recognition pipelines—especially ones that require no task-specific training data—is a key enabler for scalable educational AI. Traditional approaches rely on supervised deep learning models that require large-scale labeled training data, which is expensive and time-consuming to collect in real classroom environments [4,5].

Vision–language foundation models, exemplified by CLIP (Contrastive Language–Image Pretraining) [6], have recently emerged as powerful tools for zero-shot visual recognition. By learning joint image–text representations from web-scale data, CLIP enables classification without any task-specific training—users simply provide natural language descriptions of target categories. Several CLIP variants have since been proposed, including OpenCLIP trained on LAION-2B [7], SigLIP2 [8], EVA02-CLIP [9], and DFN-CLIP [10], each offering different training objectives, data curation strategies, or architectural innovations.

Despite their success on general benchmarks, the applicability of CLIP-family models to fine-grained classroom activity recognition remains largely unexplored. Classroom activities present unique challenges: visually similar actions (e.g., "standing" vs. "teaching"), domain-specific semantics, significant class imbalance, and inherent *multi-label* structure—a single classroom image typically contains multiple concurrent activities (e.g., a teacher simultaneously lecturing, standing, and referencing a screen). Moreover, different classroom scenes—teacher lecturing, student individual learning, subtle micro-actions, and group discussion—pose qualitatively different recognition demands that a single dataset cannot capture. The *prompt design*—how activity categories are described in natural language—can substantially affect zero-shot performance, yet systematic studies of prompt strategies for educational scenarios are scarce.

In this study, CLIP-family models are used primarily as *probes* rather than as the main methodological contribution. The main objective is to determine how prompt choice, metric choice, and backbone family change the conclusions drawn from zero-shot classroom benchmarks. Specifically, the analysis tests whether headline results remain stable once prompt wording is varied, class-balanced multi-label metrics are emphasized alongside lenient Hit@1, and multiple CLIP-family backbones are compared under a common protocol; it also examines whether TeacherBehavior remains the hardest subset because it contains semantically overlapping categories and stronger context dependence, whereas BowTurnHead and HandriseReadWrite are comparatively easier due to clearer visual boundaries.

A key observation motivating this study is that prompt-space perturbations expose an *adversarial-like vulnerability* in CLIP-family models. Unlike conventional adversarial attacks that perturb pixel values, the perturbations studied here operate entirely in the text input space: minor rephrasing of category descriptions—without any change to model weights, architecture, or image inputs—can silently collapse recognition performance by tens of percentage points. This form of *semantic perturbation* is particularly relevant to real-world behavior analysis deployments, where practitioners craft prompts without access to the test distribution and where performance degradation may go undetected if only lenient metrics such as Hit@1 are reported. Robustness under controlled input perturbations is a central concern in the security evaluation of LLMs and AI agents [11,12]; this work applies the same principle to the prompt space of vision–language models in a behavior analysis context, revealing that current CLIP-family models lack robustness to text-domain perturbations of practically trivial magnitude.

This study makes three contributions:

(1) **Metric-sensitive benchmark interpretation.** By quantifying the gap between TeacherBehavior Hit@1 (85.56%) and Macro-F1 (49.03%), supported by bootstrap

confidence intervals and paired significance tests, this study demonstrates that benchmark conclusions change materially with metric choice: Hit@1 is a misleading headline metric on multi-label educational data. Benchmark studies in this area should report class-discriminative metrics (Macro-F1, Sample-F1) and uncertainty estimates alongside Hit@1.

(2) **Prompt choice matters more than backbone choice in several settings.** Across the benchmark, prompt wording produces larger interpretive shifts than model family alone: all five models show negative CAPE delta on HandriseReadWrite, while CAPE benefits semantically overlapping subsets (TeacherBehavior, BowTurnHead). The three-principle ablation (Table 8) further shows that gains arise from visual grounding and discriminative contrast, not from prompt count, refuting the intuition that richer prompt ensembles are universally better.

(3) **Reproducible benchmark release.** Five CLIP-family backbone cached features, prompt templates (Set A/B/C), the paired-bootstrap significance script, and the three-principle ablation script are released publicly, enabling transparent reproduction of all reported results.

This paper is a *benchmarking study*: the models and CAPE are used as probes and analytical tools, not as new methods. No new training protocols, architectures, or algorithms are proposed. CAPE is a domain-specific prompt recipe designed in this study to operationalize three prompt-engineering principles for classroom activity recognition; it is not presented as a novel prompt-engineering method independent of this benchmark context.

Metric note. Throughout this paper, Hit@1 denotes multi-label top-1 accuracy: a prediction $\hat{y}$ is correct if it matches any label in the ground-truth set G_i, i.e., $\hat{y}_i \in G_i$. This is a lenient metric on multi-label data; formal definitions of all metrics appear in Section 3.5.

Scope note. This study focuses on culturally grounded classroom scenarios within the SCB5 dataset, which provides a controlled setting for analyzing prompt behavior in vision–language models. The goal is not to claim universal generalization, but to offer an empirical reference point for understanding how prompt design and evaluation protocol interact under structured, domain-specific conditions.

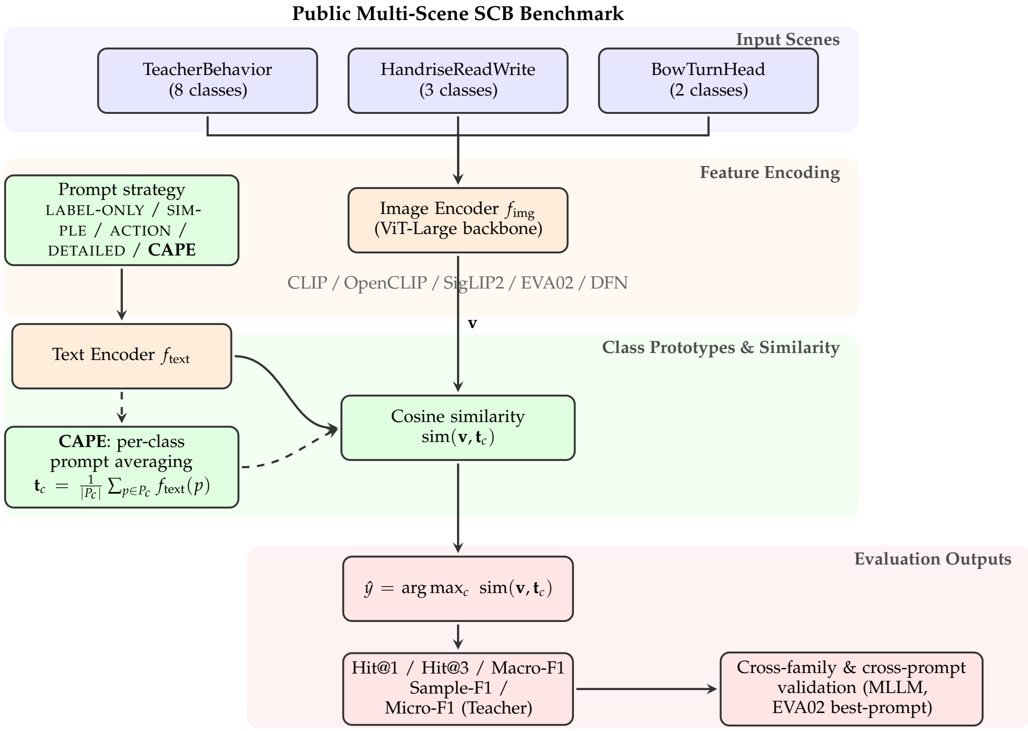


Figure 1. Overview of the evaluation pipeline. Classroom images from three public SCB sub-datasets are encoded by one of five CLIP-family vision encoders to obtain image features $\mathbf{v}$. On the text side, one of five prompt strategies is applied. For the four single-prompt strategies (LABEL-ONLY / SIMPLE / ACTION / DETAILED), the class prototype $\mathbf{t}_c$ is the text encoding of the single prompt p_c. For the **CAPE** strategy, $\mathbf{t}_c$ is the mean of the text encodings of the three class-aware prompts in P_c; the dashed-style branch through the **CAPE** block is therefore one of the five prompt strategies, not an additional module. Classification uses cosine similarity followed by argmax; evaluation reports Hit@1/Hit@3 and multi-label F1 metrics on TeacherBehavior. Cross-family and cross-prompt validation (Section 4.5.3) extends the analysis with MLLM references and an EVA02 best-prompt comparison.

2. Related Work

2.1. Classroom Activity Recognition

Classroom activity recognition has been studied through various modalities including video [13], audio [14], and sensor data [3]. Recent deep learning approaches have achieved promising results using convolutional neural networks (CNNs) and transformers for image- and video-based recognition [15,16]. However, these methods require extensive labeled data and are constrained to predefined category sets, limiting their adaptability to new educational contexts.

2.2. Vision–Language Models

CLIP [6] pioneered contrastive vision–language pretraining, enabling zero-shot transfer by matching images to natural language descriptions. OpenCLIP [17] reproduced and extended CLIP training on open datasets such as LAION-2B [7] and LAION-400M. SigLIP2 [8] builds on the sigmoid contrastive loss introduced by SigLIP [18], improving multilingual and dense feature capabilities. It was trained on the WebLI dataset. More recently, EVA02-CLIP [9] introduced masked image modeling as a pre-training auxiliary task and reported strong performance among open-source CLIP models, while DFN-CLIP [10] demonstrated that aggressive data filtering on large web-crawled corpora can substantially improve zero-shot transfer. These models have been applied to diverse domains including medical imaging [19], remote sensing [20], and action recognition [21], but their application to classroom scenarios remains limited [11].

2.3. Prompt Engineering for CLIP

The sensitivity of CLIP to prompt design has been widely documented. CoOp [22] and CoCoOp [23] learn continuous prompt embeddings to improve few-shot transfer. The original CLIP paper [6] also introduced prompt ensembling, which averages predictions across multiple handcrafted prompts to improve zero-shot transfer. More recently, CuPL [24] generates class-specific descriptions via a large language model (GPT-3) and uses them as zero-shot prompt ensembles, while WaffleCLIP [25] shows that even random descriptors concatenated to class names can match hand-crafted prompts in zero-shot transfer. Both methods share CAPE's principle of class-specific prompt aggregation, but differ in prompt source: CuPL relies on LLM-generated text, WaffleCLIP on random strings, and CAPE on manually authored, visually grounded descriptions tailored to a specific domain [12,26]. This study focuses on the zero-shot setting with manually designed prompts, systematically studying how prompt semantics affect performance in a domain-specific educational context. By including a blind-prompt control (Set C, Section 4.5.1) written without viewing dataset images—analogous to a naïve LLM prompt baseline—this study provides an empirical anchor for comparing CAPE against domain-agnostic prompt generation strategies.

3. Materials and Methods

3.1. Multi-Scene Classroom Activity Benchmark

A key contribution of this study is the construction of a multi-scene classroom activity benchmark by integrating three publicly available SCB (Smart Classroom Behavior) sub-datasets hosted on Hugging Face [27] (a fourth single-class sub-dataset, Discuss, is excluded as a degenerate case; see below). Each sub-dataset targets a distinct recognition challenge, and together they cover the full spectrum of classroom behaviors observed in Chinese K–12 and higher-education settings. The SCB resources are third-party public datasets released by wintonYF and contributors; this study only reuses these public releases for benchmarking without claiming ownership. The sub-datasets are community releases on Hugging Face; the license field on the Hugging Face repository page lists these datasets under an open license (no restrictions on research use; readers should consult the current repository page for the authoritative license text). To the best of the authors' knowledge, they have not been accompanied by a peer-reviewed publication at the time of writing; this is discussed as a limitation in Section 5.10.

As no new human-subject data were collected, no IRB approval was required under the authors' institutional guidelines; raw images are not re-distributed, no face recognition or re-identification is performed, and only aggregate model-level metrics are reported. Table 1 summarizes the benchmark.

Table 1. Overview of the multi-scene classroom activity benchmark. All sub-datasets are publicly available on Hugging Face. Train/val splits follow the original dataset release.

Sub-dataset	Train	Val	Classes	Core recognition challenge
SCB5-TeacherBehavior	8984	3240	8	Macro-level teacher postures and teaching-equipment interaction (guide, answer, on-stage interaction, blackboard-writing, teacher, stand, screen, blackboard). Multi-label: a single image typically contains 3–5 co-occurring labels.
SCB5-HandriseReadWrite	5193	1671	3	Fine-grained individual student actions (hand-raise, read, write).
SCB-BowTurnHead	1905	505	2	Subtle negative micro-actions (bow-head, turn-head) involving minimal body displacement.
Total	**16082**	**5416**	**13**	

Note. A fourth sub-dataset, SCB5-Discuss, contains 259 images labeled with a single class ("discuss"). Because any model trivially achieves 100% Hit@1 on a one-class classification task, it is excluded from all quantitative analyses and reported only as a degenerate-case remark.

TeacherBehavior.

This eight-class sub-dataset is the largest and most challenging partition. It exhibits heavy multi-label co-occurrence: "stand" appears in 97.2% and "teacher" in 92.6% of images (Table 2). A *primary label* assignment is defined only for auxiliary single-label diagnostics: each image receives the label with the smallest class index among its ground-truth annotations. This rule is deterministic but not semantically canonical, and the resulting distribution is severely skewed (Table 3). Accordingly, the main conclusions for TeacherBehavior are based on multi-label metrics, whereas primary-label results are used only for confusion analysis and controlled side-by-side comparisons with prior single-label reporting.

Table 2. Multi-label co-occurrence in the SCB5-TeacherBehavior validation set (3240 images).

Category	Images	Prevalence (%)
guide	372	11.5
answer	800	24.7
on-stage interaction	149	4.6
blackboard-writing	246	7.6
teacher	3001	92.6
stand	3148	97.2
screen	1885	58.2
blackboard	2089	64.5

Table 3. Single-label (primary) distribution for SCB5-TeacherBehavior.

Index	Category	Count	Proportion (%)
0	guide	372	11.5
1	answer	800	24.7
2	on-stage interaction	141	4.4
3	blackboard-writing	235	7.3
4	teacher	1459	45.0
5	stand	169	5.2
6	screen	24	0.7
7	blackboard	40	1.2
	Total	**3240**	**100.0**

HandriseReadWrite.

This three-class sub-dataset focuses on fine-grained student learning actions. “Hand-raise,” “read,” and “write” are visually distinct but share similar classroom backgrounds, requiring localizable action-level understanding.

BowTurnHead.

Detecting “bow-head” (looking down) and “turn-head” (looking sideways) demands recognition of extremely subtle body-displacement cues. With only 505 validation images, this sub-dataset tests whether vision–language models can capture micro-actions without task-specific training. Under the single-label primary assignment used for class-wise analysis, the validation split contains 185 bow-head images and 320 turn-head images, so the majority-class baseline is already 63.37% under single-label accuracy. The multi-label annotation file additionally credits a prediction whenever it matches *any* positive label, raising the multi-label Hit@1 majority floor to 90.69% (i.e., a classifier that always predicts “turn-head”—the class present in 458 of 505 images—achieves $458/505 = 90.69\%$ under the $\hat{y} \in G$ rule). Both floors should be kept in mind when reading high BowTurnHead numbers in Table 6; this subset is best treated as a near-saturated micro-action stress test rather than as a clean cross-model discriminator.

3.2. Models

Five representative CLIP-family models are evaluated, all using ViT-Large backbones for fair comparison (Table 4). This selection covers the major axes of variation in the CLIP landscape: training data source (proprietary vs. open), loss function (softmax vs. sigmoid contrastive), data curation strategy (unfiltered vs. filtered), and pre-training augmentation (with or without masked image modeling).

Table 4. Summary of evaluated models. All use ViT-Large backbones. SigLIP2 uses 256×256 input resolution vs. 224×224 for all others (see main text); † EVA02-L/14 architectural caveat in note below.

Model	Architecture	Training data	Key innovation	Input size
CLIP (OpenAI) [6]	ViT-L/14	WIT (400M)	Original CLIP	224
OpenCLIP (LAION) [17]	ViT-L/14	LAION-2B	Open-data reproduction	224
SigLIP2 [8]	ViT-L/16	WebLI	Sigmoid contrastive loss	256
EVA02-CLIP [9]	EVA02-L/14†	Merged-2B	Masked image modeling	224
DFN-CLIP [10]	ViT-L/14	DFN-2B	Data filtering network	224

† EVA02-L/14 adopts a masked-image-modeling (MIM) auxiliary pre-training objective and differs internally from standard contrastive ViT-L backbones despite identical input/output dimensions; cross-model comparisons involving EVA02 should be treated as indicative rather than controlled architectural ablations.

EVA02 architectural caveat. EVA02-L/14 uses a masked-image-modeling (MIM) pre-training objective that differs fundamentally from the contrastive training of the other four backbones; it is therefore architecturally non-equivalent and its results throughout this paper should be treated as indicative comparisons rather than controlled architectural ablations.

One further architectural difference warrants explicit attention: SigLIP2 operates at 256×256 pixel input resolution, versus 224×224 for all other four models. This higher input resolution may confer an implicit advantage on fine-grained spatial features (e.g., distinguishing chalk motion in blackboard-writing or subtle head orientations in BowTurnHead), and is a confounding factor in cross-model comparisons that cannot be fully controlled in this study.

3.3. Prompt Strategies

Four prompt strategies are designed with increasing semantic specificity. All prompts use human-readable class descriptions rather than raw label names (e.g., "explaining concepts while lecturing" instead of "teacher"):

(1) **Label-only**: Raw class description as the prompt, e.g., "*explaining concepts while lecturing*".
(2) **Simple**: Template "a photo of {class}", e.g., "*a photo of explaining concepts while lecturing*".
(3) **Action**: Template "a teacher is {class}", e.g., "*a teacher is explaining concepts while lecturing*".
(4) **Detailed**: Two scene-grounded templates per class, e.g., "*a classroom scene where a teacher is explaining concepts while lecturing*" and "*a teacher is explaining concepts while lecturing during a lecture*".

3.4. Class-Aware Prompt Ensemble (CAPE): A Domain-Specific Prompt Recipe

CAPE is a simple domain-specific prompt recipe that tests whether class-specific text engineering can close the gap between uniform prompts and task-specific prompts on classroom data. Standard prompt strategies apply the same template(s) uniformly across all classes. However, classroom activities have distinct visual and semantic characteristics. For example, "blackboard-writing" emphasizes chalk and hand motion, whereas "on-stage interaction" emphasizes teacher–student spatial arrangement. The hypothesis is that *class-specific* prompts can better capture these distinctions.

CAPE assigns a tailored set of $K = 3$ prompts to each class, following three design principles:

(1) **Visual grounding.** Each prompt describes observable visual evidence (e.g., "a teacher writing on the blackboard with chalk") rather than abstract concepts, anchoring the text embedding in image-level features the vision encoder can match.
(2) **Semantic diversity.** The three prompts per class cover complementary perspectives—viewpoint ("standing at the front"), action ("gesturing while talking"), and context ("interacting with students at the podium")—so that the averaged embedding spans a richer region of the shared feature space.
(3) **Discriminative contrast.** Prompts are drafted to highlight *between-class* differences rather than within-class variability. For instance, "screen" prompts emphasize digital media to distinguish from "blackboard-writing" prompts that emphasize chalk.

Importantly, CAPE prompts are designed once per class label and reused across all sub-datasets, models, and scene types without per-dataset tuning [28]. Representative examples include "a teacher writing on a blackboard with chalk" (blackboard-writing), "a teacher pointing at a projection screen" (screen), and "a student turning their head to look sideways" (turn-head). The full prompt lists are provided in the project release package.

For inference, the text embedding for class c is computed as the L2-normalized mean of all prompt embeddings assigned to that class:

$$\tilde{\mathbf{t}}_c = \frac{1}{|P_c|} \sum_{p \in P_c} f_{\text{text}}(p), \qquad \mathbf{t}_c = \ell_2(\tilde{\mathbf{t}}_c) \tag{1}$$

where P_c is the set of CAPE prompts for class c, f_{text} is the text encoder, and $\ell_2(\cdot)$ denotes L2-normalization. This matches the explicit normalization step in Algorithm 1. The predicted class for an image with L2-normalized feature $\mathbf{v}$ is:

$$\hat{y} = \arg\max_{c} \; \mathbf{v}^\top \mathbf{t}_c \tag{2}$$

where the dot product equals cosine similarity because both $\mathbf{v}$ and $\mathbf{t}_c$ are L2-normalized.

CAPE requires no training, no additional data, and no model modification—only carefully designed class-specific prompt sets. Algorithm 1 summarizes the complete inference procedure.

Algorithm 1: CAPE: Class-Aware Prompt Ensemble Inference

Input: Image set $\mathcal{X}$; class set $\mathcal{C} = \{c_1, \ldots, c_K\}$; per-class prompt sets $\{P_{c_1}, \ldots, P_{c_K}\}$; vision encoder f_{img}; text encoder f_{text}
Output: Predicted label $\hat{y}_i$ for each image x_i

`// Stage 1: Compute class-aware text embeddings (offline)`
for *each class* $c \in \mathcal{C}$ **do**
  $\tilde{\mathbf{t}}_c \leftarrow \frac{1}{|P_c|} \sum_{p \in P_c} f_{\text{text}}(p)$ `// mean-pool prompt embeddings`
  $\mathbf{t}_c \leftarrow \ell_2(\tilde{\mathbf{t}}_c)$ `// L2-normalize`
end

`// Stage 2: Zero-shot classification (per image)`
for *each image* $x_i \in \mathcal{X}$ **do**
  $\mathbf{v}_i \leftarrow \ell_2(f_{\text{img}}(x_i))$ `// encode & normalize`
  $\hat{y}_i \leftarrow \arg\max_{c \in \mathcal{C}} \; \mathbf{v}_i^\top \mathbf{t}_c$ `// cosine similarity`
end
return $\{\hat{y}_i\}$

3.5. Evaluation Protocol

Because the SCB5 dataset is inherently multi-label (each image may contain multiple activity annotations) while CLIP produces a single class prediction, a primary multi-label evaluation and an auxiliary single-label diagnostic evaluation are defined:

Multi-label Hit@1 (Hit@1).

The model's top-1 prediction is considered correct if it matches *any* ground-truth label of the image. Formally, for an image with ground-truth label set G and top-1 prediction $\hat{y}$:

$$\text{Hit@1} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[\hat{y}_i \in G_i] \tag{3}$$

This metric is lenient: it rewards the model whenever its prediction corresponds to any activity present in the scene, regardless of which specific activity it identifies.

Multi-label Hit@3 (Hit@3).

Analogously, the prediction is correct if any of the top-3 predicted classes overlaps with the ground-truth label set.

Single-label (Primary) Metrics.

For confusion matrix analysis, auxiliary macro F1, and per-class recall, each image is assigned a single *primary* label defined as the ground-truth class with the smallest index. This assignment is deterministic but introduces heavy class imbalance (Table 3), favoring lower-indexed classes (guide, answer) over higher-indexed ones (screen, blackboard). It is therefore treated as an analysis convenience rather than as a canonical target definition for TeacherBehavior.

For TeacherBehavior, the main interpretation relies on multi-label Hit@1/Hit@3 together with proper multi-label F1 metrics reported in Section 4.5.2. Primary-label metrics are retained only to expose class-collapse behavior under a forced single-label view. For the near-single-label HandriseReadWrite and BowTurnHead subsets, standard single-label accuracy and Macro-F1 remain directly interpretable.

3.6. MLLM Evaluation Protocol

The cross-family validation in Section 4.5.3 uses a deterministic single-image prompting protocol for all MLLM references. Each image is evaluated independently, and the candidate label inventory exactly matches the class set used in the CLIP-family experiments for the same subset. The user prompt asks the model to choose the single best category from the enumerated class list and to return valid JSON only, with a single integer field identifying the selected class. The same protocol is applied to TeacherBehavior, HandriseReadWrite, and BowTurnHead. This creates a fundamental output-space asymmetry on TeacherBehavior: CLIP-family models produce a full per-class similarity vector that supports both argmax and threshold-based multi-label analysis, whereas MLLMs under this protocol are constrained to a single categorical output. All TeacherBehavior cross-family comparisons should therefore be read as single-choice proxies under a shared scorer, not as mechanistically equivalent evaluations.

For the Ollama-based models used in the reported comparisons—Ollama supports a growing ecosystem of open-source LLMs such as DeepSeek [26]—requests are sent through the chat API with streaming disabled and temperature fixed at 0 so that decoding is deterministic for a given image and prompt. Internal reasoning mode is disabled, and the system instruction restricts the output to JSON. Returned predictions are parsed from the JSON field whenever possible; if malformed output occurs, a regex-based numeric fallback is applied so that the run can complete without manual intervention. The parsed label is then converted to a one-hot pseudo-logit vector and scored by the same evaluation code used for the CLIP-family models, ensuring that Hit@1, Hit@3, Macro-F1, and related summary metrics are computed by a common downstream pipeline.

4. Experiments

4.1. Experimental Setup

All experiments are conducted on four NVIDIA Tesla V100-32GB GPUs using PyTorch 2.0.1 and the OpenCLIP library [17] (version 2.24.0, CUDA 11.8). Exact package versions, model checksum hashes, and the _V100 FP16 precision configuration are recorded in `requirements.txt` and `README.md` of the release package. Models are loaded with pretrained weights and evaluated in a fully zero-shot setting (no fine-tuning). The five models are distributed across four GPUs and evaluated in parallel on all three benchmark sub-datasets. For each image, the model computes cosine similarity between the image embedding and all class text embeddings, and the class with the highest similarity is selected as the prediction.

Computational overhead of CAPE.

All five models use ViT-Large backbones (304–430M parameters; Table 4). CAPE's additional computational cost relative to single-prompt inference is confined to the text encoding stage, which is performed *once offline*: encoding $K=3$ prompts per class instead of one increases the text-side computation by $3\times$, but this is a one-time cost amounting to less than one second for all 13 classes across all three sub-datasets on a single V100 GPU. At inference time, the per-image latency is **identical** regardless of prompt strategy, because the class text embeddings $\mathbf{t}_c$ are precomputed and the only per-image operation is a single forward pass through the vision encoder followed by C dot products (where $C \leq 8$). The image encoding step dominates runtime at $\sim$8–12 ms per 224×224-pixel image on a V100-32GB. Consequently, CAPE adds negligible overhead to any deployment pipeline and is compatible with real-time classroom monitoring at typical frame rates ($\leq$30 fps).

4.2. Headline performance and prompt design

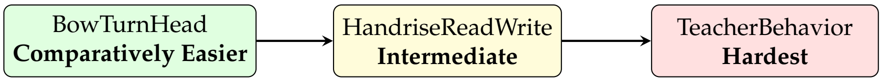
4.2.1. E1: Cross-Dataset Model Comparison

Table 6 presents the complete Hit@1 results across all five models, five prompt strategies, and three benchmark sub-datasets. The best-performing configuration per dataset is reported in Table 5.

Table 5. Best model–prompt configuration per sub-dataset (Hit@1, %). SL Acc and Macro-F1 (prim.) both use primary-label assignment; for TeacherBehavior they are auxiliary diagnostics—the corresponding multi-label Sample-F1 (60–66%) and Macro-F1 (49–60%) are reported in Table 15. The extremely low SL Acc (9.57%) and Macro-F1 (prim.) (10.07%) for TeacherBehavior reflect a collapse toward the majority class under forced single-label assignment of a fundamentally multi-label dataset—note: 9.57% SL Acc falls far below both the uniform-random baseline (12.5% for 8 classes) and the majority-class baseline (45.0%, always predicting "teacher"), confirming complete collapse to a non-discriminative prediction pattern—not poor model performance.

Sub-dataset	**Best model + prompt**	**Hit@1**	**SL Acc**†	**Macro-F1 (prim.)**†
TeacherBehavior	**SigLIP2 + CAPE**	**85.56**	**9.57**	**10.07**
HandriseReadWrite	**LAION + action**	**84.56**	**78.70**	**55.89**
BowTurnHead	**DFN + CAPE**	**93.27**	**68.51**	**53.95**

Several patterns emerge. First, **no single model dominates across all scenes**: SigLIP2 excels on TeacherBehavior, LAION leads on HandriseReadWrite, and DFN-CLIP achieves the best score on BowTurnHead. Second, **CAPE is not universally the best strategy**: on HandriseReadWrite, the best result (84.56%) comes from LAION with a simple action prompt, not CAPE. CAPE's advantage is concentrated on tasks with semantic ambiguity. Third, the HandriseReadWrite sub-dataset yields the most reliable metrics: Hit@1 (84.56%) closely tracks single-label accuracy (78.70%), confirming that multi-label inflation is minimal when annotations are near-single-label. Fourth, TeacherBehavior remains the most challenging, with the best Hit@1 reaching 85.56%. However, this value must be interpreted together with the proper multi-label results in Section 4.5.2, because the forced primary-label view collapses a fundamentally multi-label problem into an auxiliary single-label diagnostic. The corresponding subset-level ordering is summarized in Figure 2.



Empirical ordering synthesized from prompt/backbone consistency and multi-label evaluation results (Tables 5 and 15).

Figure 2. Empirical task-difficulty ladder in the SCB benchmark. CLIP-family models are treated as probes; the ordering reflects stable cross-setting behavior rather than a single model's best score.

Table 6. Complete Hit@1 (%) results: 5 models × 5 prompt strategies × 3 sub-datasets. Hit@1 follows the multi-label definition of Section 3.5 ($\hat{y}_i \in G_i$) for all three subsets, including the binary BowTurnHead subset whose annotation file stores a multi-hot positive set per image. As a context for interpretation, the multi-label majority floor (predicting the largest single class for every image) is 90.69% on BowTurnHead, 75.94% on HandriseReadWrite, and 16.42% on TeacherBehavior; the stricter single-label primary-accuracy view of the same configurations is reported in Table 5 (*SL Acc*) and the bootstrap intervals in Table 13. Bold indicates best per column.

Model	Prompt	Hit@1 (Teacher)	Hit@1 (Handrise)	Hit@1 (BowTurn)
	label-only	16.33	64.03	88.71
	simple	31.33	55.48	90.30
CLIP (OpenAI)	action	70.25	82.29	87.72
	detailed	50.86	82.41	90.69
	CAPE	84.14	67.98	89.50
	label-only	7.65	75.94	90.69
	simple	16.02	78.10	90.69
OpenCLIP (LAION)	action	24.20	**84.56**	54.46‡
	detailed	34.10	80.85	84.95
	CAPE	54.94	69.72	92.87
	label-only	16.42§	63.85	90.10
	simple	24.85	54.40	90.30
SigLIP2	action	60.68	60.62	69.11
	detailed	73.77	76.00	54.65
	CAPE	**85.56**	64.45	82.38
	label-only	77.44	82.23	63.37
	simple	78.09	81.03	63.37
EVA02-CLIP	action	78.86	60.50	64.55
	detailed	73.70	60.50	64.95
	CAPE	64.35	60.62	65.35
	label-only	12.04	84.50	64.75
	simple	35.03	82.23	57.82
DFN-CLIP	action	18.89	56.31	41.39
	detailed	43.06	78.93	44.16
	CAPE	45.86	73.73	**93.27**

‡ LAION + action on BowTurnHead collapses to 54.46%—below the label-only baseline of 90.69%—because the "a teacher is {class}" template imposes a teacher subject on student micro-gesture classes (see Section 4.2.2).
§ Equals the multi-label majority floor (16.42%), indicating degenerate prediction toward the most prevalent class ("teacher" or "stand"); label-only prompts provide no discriminative signal for this backbone on TeacherBehavior.

4.2.2. E2: Prompt Ablation

The full prompt ablation is embedded in Table 6. On the TeacherBehavior sub-dataset, the effect of prompt design is substantial: for CLIP (OpenAI), Hit@1 jumps from 16.33% (label-only) to 70.25% (action), a gain of **53.9 percentage points**. This trend is consistent across all five models, confirming that prompt semantics are the dominant factor in zero-shot classroom recognition.

Interestingly, the optimal prompt strategy *varies by scene type*. On TeacherBehavior, action and detailed prompts generally outperform simpler ones; CAPE further improves performance. On HandriseReadWrite, action prompts dominate (LAION achieves 84.56%), while CAPE is less effective—likely because the three student-action classes are already well separated by simple action descriptions. One notable anomaly: LAION + action on BowTurnHead collapses to 54.46% (‡), far below LAION's label-only baseline of 90.69%. The BowTurnHead classes ("bow" / "turn-head") are student *micro-gestures*, and the LAION action template "a teacher is {class}" imposes a teacher-subject framing that is semantically incongruent with student posture; the resulting text embedding drifts away from the visual feature space for these classes, forcing the model to predict sub-randomly between the two classes. This is a specific instance of template–subject mismatch, and reinforces the general finding that action-template prompt strategies should be verified against the subject class of each sub-dataset. On BowTurnHead, CAPE prompts provide the largest *nominal* advantage on the multi-label Hit@1 view: DFN-CLIP jumps from 64.75% (label-only) to **93.27%** with CAPE, a gain of 28.5 pp. However, the corresponding multi-label majority floor on this subset is 90.69% (Table 6 caption), so 93.27% lies only ∼2.6 pp above the multi-label majority-class baseline on the lenient metric. The stricter single-label primary-accuracy view (Table 13: 68.3% [64.6, 72.1] for DFN-CLIP+CAPE) therefore provides the more reliable evidence that CAPE genuinely helps on micro-actions. This pattern is consistent with micro-action recognition benefiting from class-specific semantic cues.

4.3. CAPE analysis and cross-model summary

4.3.1. E3: CAPE Effectiveness Across Scenes

Table 7 summarizes the CAPE gain (Hit@1) over each model's best uniform-prompt baseline on each sub-dataset. On TeacherBehavior, CAPE improves four of the five models, with gains ranging from +2.8 pp (DFN) to +20.8 pp (LAION), while EVA02-CLIP is an exception and performs better with action prompts than with CAPE. Unlike the other models, EVA02 varies only modestly across the four uniform prompts on TeacherBehavior (73.70–78.86%), suggesting relative insensitivity to wording within that prompt family; the major change appears when switching from the uniform prompts to CAPE, which drops to 64.35%. On BowTurnHead, CAPE delivers the largest single improvement of +28.5 pp for DFN-CLIP, while the EVA02 gain is only marginal (+0.4 pp). On HandriseReadWrite, CAPE underperforms the best baseline for all five models—simple or action-oriented prompts already provide strong separation among the three student-action classes. This pattern reveals a key insight: **CAPE's advantage is task-dependent**. It is most effective when categories are semantically overlapping (TeacherBehavior's 8 co-occurring labels) or involve subtle visual distinctions (BowTurnHead's micro-actions), but provides no benefit—and can even hurt—when categories are already visually well-separated (HandriseReadWrite's hand-raise vs. read vs. write).

Table 7. CAPE Hit@1 (%) vs. best baseline per model and sub-dataset. Δ denotes gain in percentage points.

	TeacherBehavior		**HandriseReadWrite**		**BowTurnHead**	
Model	CAPE	Δ	CAPE	Δ	CAPE	Δ
CLIP (OpenAI)	84.14	+13.9	67.98	−14.4	89.50	−1.2
OpenCLIP (LAION)	54.94	+20.8	69.72	−14.8	92.87	+2.2
SigLIP2	85.56	+11.8	64.45	−11.6	82.38	−7.9
EVA02-CLIP	64.35	−14.5	60.62	−21.6	65.35	+0.4
DFN-CLIP	45.86	+2.8	73.73	−10.8	93.27	+28.5

Figure 3 visualizes the CAPE gain pattern. The sign reversal on HandriseReadWrite (all five models show negative Δ) provides strong evidence that CAPE effectiveness is task-dependent.

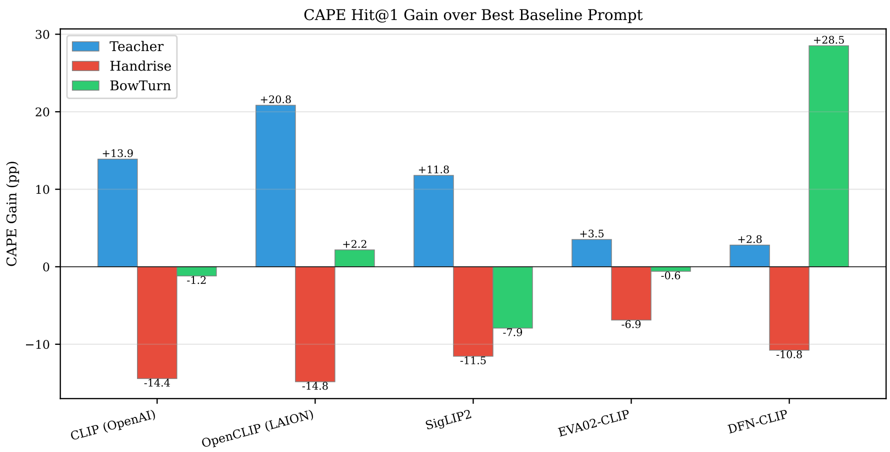


Figure 3. CAPE Hit@1 gain (pp) over each model's best baseline prompt, by sub-dataset. Positive bars indicate CAPE improvement; negative bars indicate simpler prompts outperform CAPE. The consistent negative pattern on HandriseReadWrite supports task dependency.

4.3.2. E4: Robustness of CAPE Prompt Ensembles

To assess whether CAPE's gains are robust to prompt wording rather than tied to a single handcrafted set, the prompt ensemble is perturbed along two axes: *prompt count* (using one, two, or three prompts from Set A), and *wording* (replacing Set A with independently written Set B). A mixed-pool score is also reported by averaging over five random 3-prompt draws from the union of Set A and Set B.

Set B follows the same three CAPE design principles (visual grounding, semantic diversity, and discriminative contrast) but uses independently written wording. The main benchmark fixes CAPE at $K = 3$ prompts per class, whereas this section perturbs prompt count to probe sensitivity rather than to perform per-model tuning. The verbatim Set A and Set B prompts for all thirteen classes are released alongside the code.

Robustness is strongly scene-dependent. TeacherBehavior is the least stable sub-dataset: all five models exhibit large Set A vs. Set B gaps and high mixed-prompt variance. SigLIP2 is the most extreme example, rising to 95.52% with A(2) but dropping to 31.36% with B(3). Because both values are reproduced in the archived robustness logs and JSON outputs, this swing is treated as an observed prompt-count and wording sensitivity effect rather than as a transcription artifact. In contrast, BowTurnHead is markedly more stable for several models, with mixed-prompt standard deviation as low as 1.94 pp for DFN-CLIP. HandriseReadWrite occupies a middle ground, where alternate wording often helps more than increasing prompt count.

The A(3) results should therefore be interpreted as robustness probes rather than duplicates of the fixed main setting. Small numerical differences from the main CAPE values arise because robustness uses an independently executed script. The full robustness matrix is provided in the release package to keep the main text concise.

Three-principle ablation.

To attribute CAPE's gains to its design principles rather than to arbitrary prompt choices, each of the three CAPE design principles is individually degraded while keeping the other two and the prompt count ($K = 3$) fixed: (C1) drop *visual grounding* by replacing concrete physical descriptions with abstract conceptual phrasings; (C2) drop *semantic diversity* by using three near-paraphrases of one viewpoint; (C3) drop *discriminative contrast* by using generic per-class wording that does not separate classes (e.g., "a classroom scene with a student" for all student-action classes). Table 8 reports zero-shot results on TeacherBehavior (multi-label) and BowTurnHead (single-label) for the three backbones with cached features; HandriseReadWrite is summarized in the release package because the small label set leaves little room for principle separation.

On TeacherBehavior, dropping visual grounding causes the most consistent damage to Sample-F1 (CLIP/OpenAI -17.6 pp; SigLIP2 -14.6 pp) and almost completely collapses Hit@1 ($< 10\%$ for all three backbones, indicating that abstract prompts no longer cluster by class identity). For DFN-CLIP, the visual-grounding ablation barely changes Sample-F1 ($49.92 \to 50.23$) but still collapses Hit@1 from 45.90 to 5.68%; this is consistent with DFN-CLIP's TeacherBehavior CAPE baseline being already weak (45.86% Hit@1, Table 6)—i.e., DFN's text encoder is misaligned with the teacher-behavior label semantics for this prompt family rather than near a ceiling. Dropping discriminative contrast also reduces Sample-F1 substantially (CLIP -16.7 pp; SigLIP2 -10.6 pp) and collapses BowTurnHead single-label primary accuracy to the degenerate single-class floor of 36.63% across all three backbones (the minority-class fraction reached when argmax ties drive every sample to a single fixed class), the cleanest signature of indistinguishable per-class prompts. In contrast, dropping only semantic diversity yields modest changes ($+2.3$ to -4.2 pp Sample-F1 on TeacherBehavior); for DFN-CLIP, both C2 and C3 ablations even *slightly* raise the oracle Sample-F1 above CAPE-Full ($49.92 \to 51.05/54.94$). This is not a refutation of CAPE but a useful negative signal: when the backbone's text encoder is already misaligned with a label family on a given subset, adding more class-anchored prompt structure is not free—it can mildly hurt. CAPE should therefore be reported with per-backbone calibration rather than asserted as universally beneficial. Overall, the pattern reveals that CAPE's gains over a one-line baseline arise primarily from visual grounding and discriminative contrast, with semantic diversity acting as a secondary stabilizer.

Oracle vs. non-oracle multi-label F1.

Sample-F1, Macro-F1, and Micro-F1 in Table 8 use the standard CAPE-paper convention of predicting the top-k classes per image where k equals the ground-truth positive count for that image. This "oracle-k" convention informs the model how many labels to emit and is therefore an upper bound on what a fixed-threshold zero-shot system would achieve on the same logits. The headline LP-vs.-CAPE comparison in Table 11 uses the stricter non-oracle setting (CAPE multi-label F1 is taken from Table 15, which selects a fixed cosine threshold per backbone without oracle knowledge of k); under that protocol the LP advantage is even larger. Readers should therefore not directly compare Table 8 Sample-F1 numbers to Table 11 Sample-F1 numbers, and should treat the former as relative ablation deltas rather than as absolute deployable F1.

Table 8. CAPE three-principle ablation. C0 is the original CAPE-A. Each ablation degrades one principle while holding the other two and prompt count ($K = 3$) constant. Multi-label F1 uses oracle-k (upper bound; see paragraph above for details)—do not compare directly to LP F1 in Table 11. Hit@1 on BowTurnHead is single-label primary accuracy (505 images; majority floors: SL 63.37%, ML 90.69%). Italic: degenerate floor (36.63%). Bold: strongest ablation contrast per row.

		TeacherBehavior				BowTurnHead
Backbone	**Condition**	Hit@1	Sample-F1	Macro-F1	Micro-F1	Hit@1
CLIP (OpenAI)	C0 CAPE-Full	84.32	**66.57**	54.19	69.63	**63.56**
	C1 ¬VisGround	7.59	48.94	40.95	53.66	44.16
	C2 ¬SemDiv	85.80	68.89	54.79	72.23	45.94
	C3 ¬Discrim	56.82	49.85	48.06	54.63	*36.63*
SigLIP2	C0 CAPE-Full	85.49	**64.97**	56.67	68.55	**64.75**
	C1 ¬VisGround	7.13	50.39	42.57	55.58	40.20
	C2 ¬SemDiv	79.38	60.76	56.11	64.33	56.24
	C3 ¬Discrim	68.64	54.32	51.42	58.57	*36.63*
DFN-CLIP	C0 CAPE-Full	45.90	49.92	52.72	56.26	**68.32**
	C1 ¬VisGround	5.68	50.23	38.97	54.53	40.20
	C2 ¬SemDiv	37.62	51.05	48.39	56.17	41.78
	C3 ¬Discrim	68.02	54.94	51.52	59.44	*36.63*

4.3.3. E5: Descriptive Cross-Model Summary

To provide a compact descriptive summary, each model's best Hit@1 (across all prompts) per sub-dataset is listed in Table 9. Because the three subsets differ in label structure, annotation leniency, and class count, **no cross-subset average is computed**; per-subset columns are provided for reference only, and no cross-subset ordering of backbones is implied.

Table 9. Best per-subset Hit@1 (%) for each model (best prompt selected post hoc per model per subset; post hoc selection inflates Hit@1 relative to any pre-committed single-prompt deployment, so values represent an optimistic upper bound on per-subset performance). No average across subsets is reported because the subsets have fundamentally different label structures (8-class multi-label vs. 3-class near-single-label vs. 2-class binary) and Hit@1 leniency; any cross-subset average would conflate incomparable quantities. Selected best-prompt strategies are CLIP (OpenAI): Teacher=CAPE, Handrise=detailed, BowTurn=detailed; SigLIP2: Teacher=CAPE, Handrise=detailed, BowTurn=simple; OpenCLIP (LAION): Teacher=CAPE, Handrise=action, BowTurn=CAPE; EVA02-CLIP: Teacher=action, Handrise=label-only, BowTurn=CAPE; DFN-CLIP: Teacher=CAPE, Handrise=label-only, BowTurn=CAPE.

Model	**Teacher**	**Handrise**	**BowTurn**
CLIP (OpenAI)	84.14	82.41	90.69
SigLIP2	**85.56**	76.00	90.30
OpenCLIP (LAION)	54.94	**84.56**	92.87
EVA02-CLIP	78.86	82.23	65.35
DFN-CLIP	45.86	84.50	**93.27**

4.4. Reliability and supervised baselines

4.4.1. E6: Error Analysis and Class-Level Behavior

To move beyond aggregate Hit@1, per-class prediction patterns are analyzed using confusion matrices (Figure 4) and per-class recall (Figure 5).

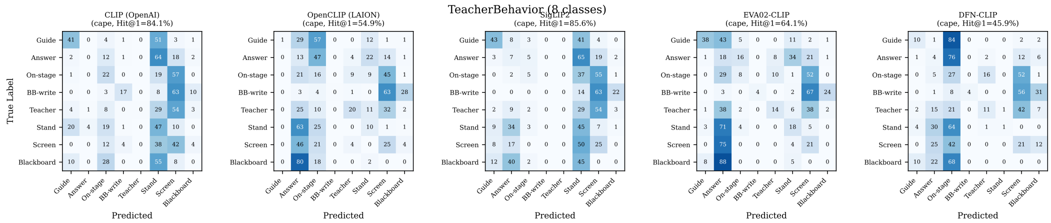


Figure 4. Normalized confusion matrices (row = true, column = predicted, values in %) for TeacherBehavior under each model's best prompt. All five models heavily over-predict "teacher" and "stand," which together account for >90% of predictions.

The confusion matrices reveal a consistent failure mode: all five models collapse predictions toward the "teacher" and "stand" classes, which have the highest base rates (45.0% and 5.2% of labels, respectively, but appear in >97% of images as co-occurring labels). Minority classes such as "screen" (0.7%) and "blackboard" (1.2%) are almost never predicted, explaining the <14.5% macro F1 despite >84% Hit@1.

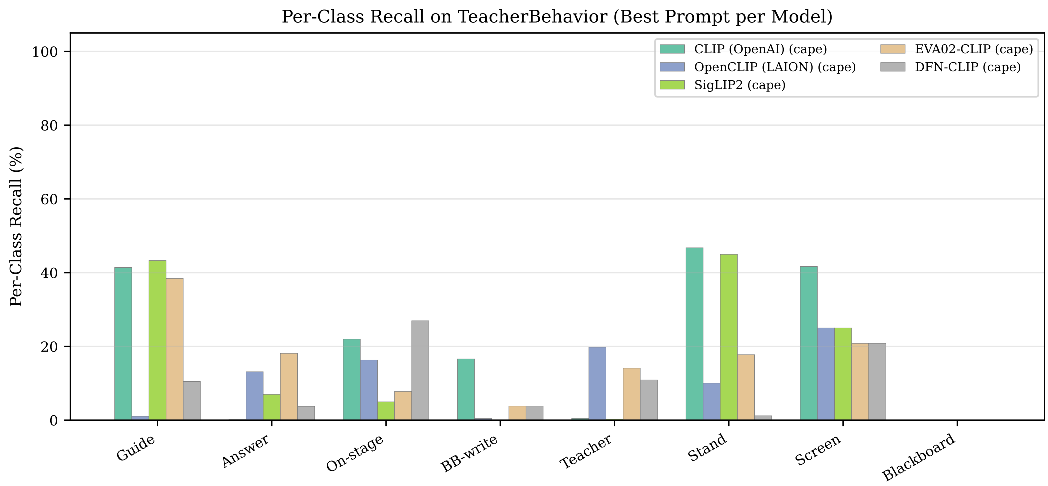


Figure 5. Per-class recall (%) on TeacherBehavior for each model (best prompt). "Teacher" and "stand" achieve high recall; "screen" and "blackboard" are near zero.

Figure 5 quantifies this imbalance: "teacher" recall exceeds 60% for four of five models under their best prompt, while "screen" recall is 0% for all models except SigLIP2 (which achieves ∼4% via detailed prompts). This is not a model failure *per se*, but a consequence of multi-label structure: "teacher" appears as a positive co-occurring label in >92% of TeacherBehavior images, so any model predicting "teacher" achieves high Hit@1; by contrast, "screen" appears in only 0.7% of images (primary-label count) and is almost never a correct top-1 prediction. Zero-shot models follow the path of least resistance in the shared embedding space.

Figure 6 further illustrates the prediction distribution mismatch.

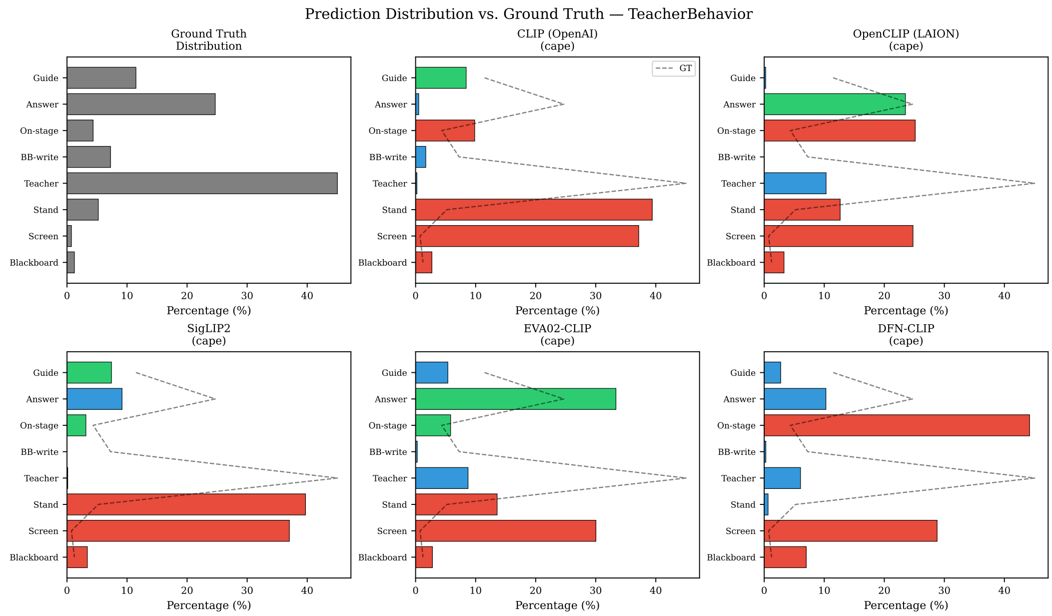


Figure 6. Prediction distribution (colored bars) vs. ground-truth distribution (dashed line) for TeacherBehavior. Red indicates over-prediction; blue indicates under-prediction relative to the true distribution.

The heatmap in Figure 7 provides a compact overview of the full 5 model $\times$ 5 prompt interaction across all three sub-datasets. 492 493

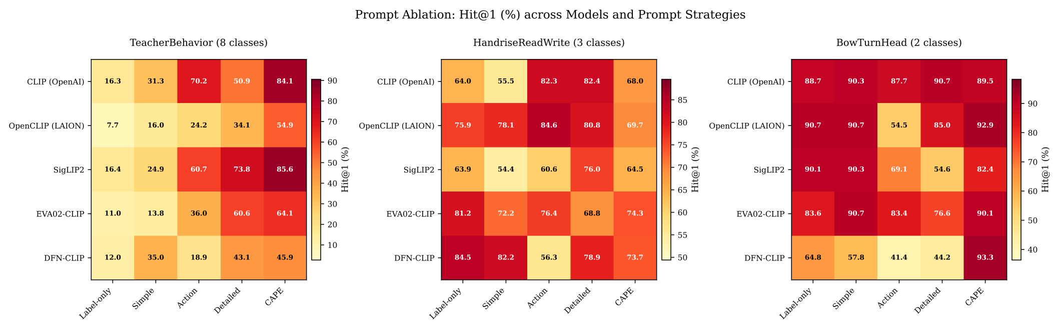


Figure 7. Hit@1 (%) heatmap across five models and five prompt strategies for each sub-dataset. The strong column effect (prompt strategy) on TeacherBehavior and the strong row effect (model choice) on HandriseReadWrite are clearly visible.

Two patterns emerge from the heatmap: (1) On TeacherBehavior, the dominant gradient runs left-to-right (prompt complexity), confirming that prompt design is the primary factor. (2) On HandriseReadWrite, the pattern is more model-dependent, with LAION and DFN achieving high scores across multiple prompt strategies while SigLIP2 is more prompt-sensitive. This corroborates the task-dependent nature of both prompt and model selection. 494 495 496 497 498 499

4.4.2. E7: Supervised Baseline via Linear Probing 500

To contextualize zero-shot performance, a supervised baseline is established by training a logistic regression classifier on frozen CLIP image features. For each backbone, image features are extracted with the frozen ViT-Large vision encoder *without* L2-renormalisation across train/val (the encoder's output convention is preserved as released), then standardised with a per-feature `StandardScaler` fitted on the training split only and applied to validation. A multinomial logistic regression is fitted on this standardised representation with L2 regularisation (`scikit-learn` default `LogisticRegression(C=1.0, max_iter=1000,` 501 502 503 504 505 506 507 `solver="lbfgs"`), no class re-weighting, no validation-set hyperparameter search) on the official training split provided by the dataset release ($N = 8984$ images for TeacherBehavior, $\approx$3500 for HandriseReadWrite, $\approx$2050 for BowTurnHead). For HandriseReadWrite and BowTurnHead, this single multinomial model yields a per-image class index, and accuracy is reported on the validation split. For TeacherBehavior, an additional set of *per-class binary classifiers* (one-vs-rest `LogisticRegression` with the same hyperparameters) is trained for multi-label evaluation; binary predictions use the default decision boundary $p \geq 0.5$, and Sample-F1, Macro-F1, and Micro-F1 are reported in Table 11 without per-class threshold tuning on the validation split. No image augmentation, no feature fine-tuning, and no validation-set early stopping is used—the comparison isolates the value of labeled data alone and makes the LP numbers directly reproducible from the released cached features.

Table 10 presents the results. Two findings are observed. First, on single-label metrics, the supervised advantage is clear: on HandriseReadWrite, linear probing reaches 83.18% accuracy (CLIP) versus 67.98% for CAPE—a +15.2 pp supervised advantage. On BowTurnHead, models are roughly comparable.

Second, the TeacherBehavior comparison requires nuance given the multi-label structure of the dataset and the potential for metric distortion. The linear probe trained on primary labels achieves 71–77% Hit@1 versus 46–86% for zero-shot CAPE. However, when evaluated with proper multi-label metrics using per-class binary classifiers, the linear probe achieves Sample-F1 of 88–90%, Macro-F1 of 68–78%, and Micro-F1 of 89–91% (Table 11), substantially outperforming zero-shot CAPE (Sample-F1 60–66%, Macro-F1 49–61%, Micro-F1 61–70% from Table 15). These results confirm that **the zero-shot advantage on Hit@1 largely reflects lenient multi-label score inflation**, not a genuine superiority over supervised learning. Under the same multi-label protocol, supervised linear probing with 8984 labeled training images remains stronger across all five backbones.

Table 10. Linear probe (LP) vs. zero-shot CAPE: single-label comparison. LP trains a logistic regression on frozen ViT-L features using primary labels. Hit@1 uses multi-label evaluation; Acc and Macro-F1 use primary-label assignment. Best per dataset in bold.

	TeacherBehavior		**HandriseReadWrite**		**BowTurnHead**	
Model backbone	LP Hit@1	ZS Hit@1	LP Acc	ZS Hit@1	LP Acc	ZS Hit@1
CLIP (OpenAI)	77.10	**84.14**	**83.18**	67.98	63.96	**89.50**
OpenCLIP (LAION)	76.85	54.94	**80.49**	69.72	66.73	**92.87**
SigLIP2	76.33	**85.56**	**73.19**	64.45	67.33	**82.38**
EVA02-CLIP	75.77	64.35	**76.78**	60.62	59.21	**65.35**
DFN-CLIP	71.98	45.86	**80.61**	73.73	60.20	**93.27**

Table 11. Multi-label comparison on TeacherBehavior: supervised LP (per-class binary classifiers trained on 8984 images) vs. zero-shot CAPE (from Table 15). LP achieves substantially higher multi-label F1 across all backbones, confirming that the zero-shot Hit@1 advantage in Table 10 reflects the lenient multi-label evaluation criterion rather than a genuine gap.

	Supervised LP (multi-label)				**Zero-Shot CAPE (multi-label)**			
Backbone	Hit@1	S-F1	Ma-F1	Mi-F1	Hit@1	S-F1	Ma-F1	Mi-F1
CLIP	**77.10**	88.61	**78.42**	90.64	84.14	64.19	55.62	66.38
LAION	76.85	88.83	75.74	90.55	54.94	65.20	57.04	67.38
SigLIP2	76.33	87.53	68.02	89.50	**85.56**	59.94	49.03	61.19
EVA02	75.77	**89.62**	77.31	**91.02**	64.35	**66.46**	**60.50**	**69.99**
DFN	71.98	88.52	73.60	90.42	45.86	63.59	57.07	66.16

S-F1 = Sample-F1, Ma-F1 = Macro-F1, Mi-F1 = Micro-F1. Bold marks the best backbone per metric column. LP outperforms zero-shot CAPE on every F1 metric for every backbone, while zero-shot Hit@1 occasionally exceeds LP Hit@1 because Hit@1 uses the lenient multi-label criterion ($\hat{y} \in G$) whereas LP is trained against primary labels.

4.4.3. E8: Bootstrap Confidence Intervals

To quantify statistical reliability on small datasets, 95% bootstrap confidence intervals (1000 resamples with replacement) are computed for the CAPE zero-shot Hit@1 setting on all sub-datasets. Table 13 reports the intervals. These intervals quantify uncertainty for the CAPE configuration only; they are not formal significance tests for the post hoc best-prompt averages in Table 9.

On BowTurnHead (505 images), the bootstrap intervals are reported on the stricter *single-label primary accuracy* (the majority floor of which is 63.37%), and the 95% CI widths range from 7 to 9 pp with substantial overlap between models. For example, CLIP achieves 63.4% [59.4, 67.3] while DFN-CLIP achieves 68.3% [64.6, 72.1]—the intervals overlap, indicating **no statistically significant difference** at the 95% level. The corresponding multi-label Hit@1 figures in Table 6 (89.50% / 93.27%) sit just $\sim$0–3 pp above the multi-label majority floor of 90.69%; **the headline ranking on this subset should therefore be read as a small effect on top of a near-saturated lenient metric**, not as a strong cross-model gap. Similarly, on TeacherBehavior, SigLIP2 (85.5% [84.4, 86.7]) and CLIP (84.1% [83.0, 85.4]) show overlapping CIs despite a nominal 1.4 pp gap.

On HandriseReadWrite, CIs are narrower ($\sim$5 pp width) due to the larger sample size, and model differences are somewhat more reliable, though all CAPE Hit@1 values cluster between 56–61%.

Paired-bootstrap significance test.

Marginal CIs are conservative for cross-model comparison because the same images are scored by every backbone. Table 12 therefore reports a *paired* bootstrap (5000 resamples with replacement, two-sided, seed 42) on the per-sample CAPE Hit@1 indicators, computed directly from the released CAPE logits caches (`data/feature_cache/`) for the three backbones with full-coverage caches (CLIP/OpenAI, DFN-CLIP, SigLIP2). OpenCLIP (LAION) and EVA02-CLIP are excluded from the paired test because their per-sample CAPE logits caches were not fully cached across all three sub-datasets at the time of this release. Their marginal 95% CIs are still reported in Table 13 and support the same qualitative conclusions. All five models participate in the marginal bootstrap analysis; the paired test is therefore an illustrative complement to—not a replacement for—the marginal CIs. Three observations follow. (i) On TeacherBehavior, OpenAI vs. SigLIP2 is *not* significant ($p = 0.089$), confirming the visual CI overlap; both are clearly different from DFN-CLIP ($p < 10^{-3}$). (ii) On BowTurnHead, OpenAI vs. DFN is paired-significant ($\Delta = -4.75$ pp, 95% CI $[-7.5, -2.2]$, $p < 10^{-3}$) even though the marginal CIs in Table 13 overlap, illustrating that overlapping marginal CIs do not imply paired non-significance. (iii) On HandriseReadWrite, OpenAI vs. SigLIP2 is significant ($p = 0.006$) but the effect size is small ($\Delta = 1.7$ pp). These results refine the conservative ranking statements made elsewhere in the paper without changing any qualitative conclusion.

Table 12. Paired-bootstrap significance test for CAPE per-sample Hit@1, computed on the cached `logits_cape` fields. "Acc" columns are point estimates from the same caches (multi-label argmax for TeacherBehavior; single-label primary accuracy for HandriseReadWrite/BowTurnHead, identical in definition to Table 13). $\Delta = \text{Acc}_A - \text{Acc}_B$ in pp; 95% CI is from 5000 paired bootstrap resamples; p is two-sided. Pairs are restricted to backbones with full-coverage feature caches.

Subset	**A**	**B**	Acc_A	Acc_B	Δ (pp)	95% CI (pp)	p
Teacher	OpenAI	DFN	84.32	45.90	$+38.43$	$[+36.6, +40.3]$	$< 10^{-3}$
Teacher	OpenAI	SigLIP2	84.32	85.49	-1.17	$[-2.5, +0.2]$	0.089
Teacher	DFN	SigLIP2	45.90	85.49	-39.60	$[-41.4, -38.0]$	$< 10^{-3}$
Handrise	OpenAI	DFN	57.99	56.91	$+1.08$	$[-0.1, +2.3]$	0.089
Handrise	OpenAI	SigLIP2	57.99	56.25	$+1.74$	$[+0.5, +3.0]$	0.006
Handrise	DFN	SigLIP2	56.91	56.25	$+0.66$	$[-0.5, +1.9]$	0.306
BowTurn	OpenAI	DFN	63.56	68.32	-4.75	$[-7.5, -2.2]$	$< 10^{-3}$
BowTurn	OpenAI	SigLIP2	63.56	64.75	-1.19	$[-5.7, +3.4]$	0.635
BowTurn	DFN	SigLIP2	68.32	64.75	$+3.56$	$[-0.6, +7.7]$	0.104

Table 13. Bootstrap 95% confidence intervals for CAPE zero-shot top-1 performance (%). Mean [lower, upper] from 1000 bootstrap resamples. **Metric note:** On TeacherBehavior the value is multi-label Hit@1 ($\hat{y}_i \in G_i$, identical in definition to Table 6); on HandriseReadWrite and BowTurnHead the value is single-label primary accuracy ($\hat{y}_i = y_i^{\text{primary}}$), which is stricter than the multi-label Hit@1 reported in Table 6. As a reference, the multi-label majority floor on BowTurnHead is 90.69% (any prediction landing on the larger label set "turn-head" counts as a hit); the corresponding single-label primary-accuracy majority floor is 63.37%.

Model	**Hit@1 (ML, Teacher)**	**Accuracy (SL, Handrise)**	**Accuracy (SL, BowTurn)**
CLIP (OpenAI)	84.1 [83.0, 85.4]	58.0 [55.7, 60.4]	63.4 [59.4, 67.3]
OpenCLIP (LAION)	55.0 [53.4, 56.8]	60.9 [58.5, 63.1]	67.7 [63.8, 71.5]
SigLIP2	85.5 [84.4, 86.7]	56.3 [53.9, 58.7]	64.7 [60.6, 68.9]
EVA02-CLIP	64.4 [62.8, 66.1]	60.7 [58.3, 63.0]	65.3 [61.2, 69.3]
DFN-CLIP	45.9 [44.2, 47.5]	57.0 [54.6, 59.3]	68.3 [64.6, 72.1]

4.5. Independent validation and extensions

4.5.1. E9: Independent Prompt Validation (Blind Set C)

A potential concern with CAPE is unconscious overfitting: the same authors design and evaluate prompts on the same test set. To assess this, a *blind* prompt set (Set C) is constructed, written solely from class names without examining any dataset images. Set C simulates what a third party—or a language model—would produce given only the category labels and no domain knowledge of Chinese classroom footage.

Table 14 compares Hit@1 across three prompt sets: Set A (original CAPE), Set B (alternate wording by the same authors), and Set C (blind).

Table 14. Hit@1 and Macro-F1 (%) comparison of CAPE Set A (original), Set B (alternate wording), and Set C (blind, written without viewing dataset images). Macro-F1 uses primary-label assignment for TeacherBehavior and standard single-label assignment for the other two sub-datasets.

	TeacherBehavior			HandriseReadWrite			BowTurnHead		
Model	A	B	C	A	B	C	A	B	C
Hit@1									
CLIP	84.1	72.9	91.9	58.0	70.6	36.8	63.4	54.3	38.0
LAION	55.0	35.4	92.2	60.8	63.6	61.0	67.7	61.2	53.1
SigLIP2	85.5	31.4	91.7	56.3	70.2	65.1	64.7	48.3	67.1
EVA02	64.4	39.2	92.1	60.6	71.2	56.9	65.3	53.7	53.7
DFN	45.9	22.2	92.4	56.9	72.4	46.3	68.3	62.2	58.2
Macro-F1 (primary-label for TeacherBehavior; single-label accuracy-based for Handrise and BowTurn)									
CLIP	12.7	10.9	10.4	38.8	53.0	32.8	43.4	50.0	29.7
LAION	7.5	12.8	10.3	42.1	45.2	51.3	51.2	60.2	53.1
SigLIP2	10.0	10.6	11.9	34.8	51.9	54.2	59.8	48.2	51.8
EVA02	14.4	13.6	10.9	43.7	51.0	47.3	47.9	53.6	53.5
DFN	7.1	10.8	12.3	38.2	52.2	37.7	53.5	60.4	54.7

The data show a methodological issue. On TeacherBehavior, Set C achieves ∼92% Hit@1 across all models—far above Set A (45–86%). However, this apparent superiority is a *label-prevalence bias under multi-label evaluation*. Set C's generic prompts (e.g., "a person in the role of a teacher or instructor" for class "teacher") systematically predict the two most prevalent labels ("teacher" at 92.6% and "stand" at 97.2%), which appear in nearly every image. Because Hit@1 succeeds when the prediction matches *any* ground-truth label, Set C achieves high Hit@1 by exploiting label co-occurrence rather than discriminating between classes. This is confirmed by the Macro-F1 scores: Set C achieves only 10–12% Macro-F1 on TeacherBehavior, comparable to or below Set A's values, demonstrating limited ability to distinguish minority classes.

Conversely, on HandriseReadWrite and BowTurnHead—where multi-label inflation is absent—Set C is usually weaker than the domain-aware prompts and does not show a stable advantage across models or metrics, confirming that **domain-aware prompt design (CAPE) provides genuine value for fine-grained activity discrimination**. The blind set lacks the visual grounding and discriminative contrast that CAPE's design principles provide.

This analysis also serves as a cautionary note: **Hit@1 alone is insufficient for evaluating prompts on multi-label data.** Future work should pair Hit@1 with class-discriminative metrics (Macro-F1, per-class recall) to avoid rewarding degenerate predictions that exploit label co-occurrence.

4.5.2. E10: Multi-Label Evaluation

Because primary-label evaluation is lenient on multi-label data, proper multi-label metrics on TeacherBehavior are computed using two approaches: (1) thresholding cosine similarity scores at various cutoffs to produce multi-label predictions, and (2) using top-k predictions as surrogate multi-label outputs. Table 15 reports the best configuration per model.

Table 15. Multi-label evaluation on TeacherBehavior (CAPE zero-shot). For each model the strategy (fixed cosine threshold τ or fixed top-k) was selected by Sample-F1 on the validation set; the chosen strategy is therefore *oracle-tuned* and represents an upper bound for a blind fixed-threshold deployment. Three of the five models (CLIP, OpenCLIP, EVA02) converge on the same $\tau = 0.20$ threshold, making their rows directly comparable at a common operating point; SigLIP2 (top-3) and DFN-CLIP ($\tau = 0.15$) use different oracle-selected strategies.

Model	Strategy	Sample-F1	Macro-F1	Micro-F1
CLIP (OpenAI)	$\tau = 0.20$	64.19	55.62	66.38
OpenCLIP (LAION)	$\tau = 0.20$	65.20	57.04	67.38
SigLIP2	top-3	59.94	49.03	61.19
EVA02-CLIP	$\tau = 0.20$	**66.46**	**60.50**	**69.99**
DFN-CLIP	$\tau = 0.15$	63.59	57.07	66.16

At the optimal threshold ($\tau = 0.20$), most models achieve Sample-F1 of 64–66% and Macro-F1 of 55–60%. EVA02-CLIP achieves the highest multi-label Macro-F1 (60.50%) despite ranking only third in Hit@1 (64.35%)—an inversion that highlights the complementary nature of the two metrics. SigLIP2, which tops the Hit@1 ranking (85.56%), achieves only 49.03% multi-label Macro-F1, revealing that its high Hit@1 is partly driven by predicting high-prevalence labels rather than by genuinely discriminating across all eight categories.

These multi-label metrics provide a more faithful picture of zero-shot model capabilities on inherently multi-label educational data and complement Hit@1 in benchmarking studies.

4.5.3. E11: Cross-Family and Cross-Prompt Validation

Two cross-family analyses examine the generality of the main conclusions beyond CLIP-family encoders: (1) a comparison against contemporary MLLMs on the same three SCB5 subsets, and (2) a direct comparison between EVA02-CLIP under CAPE and EVA02-CLIP under its best prompt strategy after checking all five prompt options. The MLLM comparison uses three Ollama-distributed models, `qwen3.5:9b`, `qwen3.5:27b`, and `gemma4:26b`, in a zero-shot setting and reports both Hit@1 and Macro-F1 under the deterministic protocol defined in Section 3.6.[1]

Table 16. Cross-family validation: MLLM comparison against the best zero-shot CLIP reference. "Best CLIP reference" denotes the strongest CLIP-family configuration on each subset (TeacherBehavior: SigLIP2+CAPE; HandriseReadWrite: OpenCLIP+action; BowTurnHead: DFN-CLIP+CAPE). Values are percentages. Weighted metrics are sample-size-weighted over the three subsets ($N = 5416$). † **TeacherBehavior MLLM Hit@1 is not directly comparable to CLIP**: MLLMs are constrained to single-label output while CLIP exposes a full similarity vector (Section 3.6); the comparison should be read as a single-choice proxy under a shared scorer, not as an exact mechanistic match. * Hit@1 under TeacherBehavior: MLLM rows use **single-label forced choice** (SL); the CLIP row uses the standard multi-label definition ($\hat{y} \in G$).

	TeacherBehavior†		HandriseReadWrite		BowTurnHead		Weighted	
Model	Hit@1*	Macro-F1	Hit@1	Macro-F1	Hit@1	Macro-F1	Hit@1	Macro-F1
Best CLIP reference	85.56	10.07	84.56	55.89	93.27	53.95	85.97	28.30
`qwen3.5:9b` (Ollama)	74.38	15.89	50.33	43.52	52.67	52.13	64.94	27.79
`qwen3.5:27b` (Ollama)	87.78	18.50	54.10	47.24	63.17	62.34	75.09	31.46
`gemma4:26b` (Ollama)	85.40	20.05	63.38	52.85	63.37	63.11	76.55	34.18

[1] The names `qwen3.5:Xb` and `gemma4:Yb` refer to local Ollama-distribution tags actually deployed in this study; they are not identical to upstream Hugging Face release names of the Qwen and Gemma series, and exact tag identifiers and quantization are recorded in the release package for reproducibility.

Table 16 refines the zero-shot picture in two ways. First, MLLMs do not remove task dependency: the best CLIP reference remains clearly stronger on HandriseReadWrite by Hit@1 (84.56% vs. 63.38% for `gemma4:26b`) and remains strongest on BowTurnHead by Hit@1. Second, class-balanced metrics alter the interpretation of teacher behavior. Although `qwen3.5:27b` and `gemma4:26b` reach 87.78% and 85.40% Hit@1, their Macro-F1 values remain only 18.50% and 20.05%, confirming that high top-1 performance on this multi-label subset does not imply robust class discrimination. On TeacherBehavior, this comparison should also be interpreted with the protocol asymmetry from Section 3.6 in mind: MLLMs are restricted to one label, whereas CLIP-family models expose a full score vector. On BowTurnHead, by contrast, MLLMs achieve higher Macro-F1 (62–63%) than the best CLIP reference (53.95%), despite lower Hit@1, suggesting more balanced binary discrimination but weaker top-1 precision under the current prompt protocol.

Table 17. EVA02 CAPE vs. best-prompt results (Hit@1, %). The table contrasts CAPE results with the best result obtained after checking all five prompt strategies under the unified evaluation pipeline.

Subset	CAPE	Best prompt strategy	Best Hit@1	Gain
TeacherBehavior	64.35	action	78.86	+14.51
HandriseReadWrite	60.62	label-only	82.23	+21.61
BowTurnHead	65.35	cape	65.35	0.00

The EVA02 CAPE vs. best-prompt comparison in Table 17 shows that prompt choice materially affects the interpretation of this backbone. On TeacherBehavior, EVA02 reaches 64.35% under CAPE, but rises to 78.86% when the best prompt strategy is used; that best strategy is action. On HandriseReadWrite, EVA02 also benefits strongly from prompt matching, improving from 60.62% with CAPE to 82.23% with label-only prompts. On BowTurnHead, by contrast, CAPE is already the best prompt strategy at 65.35%, so checking the other strategies produces no gain. This pattern confirms that prompt-strategy disclosure is necessary for fair cross-model comparison and that reporting only one fixed strategy can misstate a backbone's relative standing.

4.5.4. E12: Comparison with Generic Prompt-Engineering Baselines

CAPE is a domain-specific instantiation of CLIP prompt engineering and is not claimed as a methodological contribution beyond the existing prompt-engineering literature. To make this position concrete, two representative literature baselines are evaluated under an otherwise identical pipeline (same image features from `data/feature_cache/`, same Hit@1 protocol, only the text-side prompts vary): **CuPL** [24], which builds the per-class text prototype as the mean text embedding of ∼25 short LLM-generated descriptions (here generated by a local `gemma4:26b` model with five descriptive query templates per class), and **WaffleCLIP** [25], which ensembles 16 random-token-suffix prompts per class to dilute template bias. A single-template “`a photo of {cls}`” prompt (PLAIN) is included as a sanity floor. Verbatim prompt files for all four strategies are released alongside the experiment script.

Table 18. Hit@1 (%) of CAPE versus generic prompt-engineering baselines (PLAIN single template, CUPL LLM-generated descriptions, WAFFLECLIP random-suffix ensemble) on the same image-feature caches. The strongest entry per row is in bold; ties at the second decimal are both bolded. Multi-label TeacherBehavior uses the same argmax-then-membership Hit@1 as Table 6.

Backbone	**Subset**	PLAIN	CAPE	WAFFLECLIP	CUPL
CLIP (OpenAI)	TeacherBehavior	72.41	**84.17**	57.93	36.64
CLIP (OpenAI)	HandriseReadWrite	55.06	**57.99**	55.48	57.93
CLIP (OpenAI)	BowTurnHead	60.59	**63.76**	60.20	**63.76**
DFN-CLIP	TeacherBehavior	**68.61**	45.90	63.89	42.65
DFN-CLIP	HandriseReadWrite	56.49	56.91	**58.65**	55.60
DFN-CLIP	BowTurnHead	64.55	**68.32**	66.34	63.37
SigLIP2	TeacherBehavior	82.87	**85.49**	72.87	72.41
SigLIP2	HandriseReadWrite	55.72	56.25	55.06	**61.82**
SigLIP2	BowTurnHead	56.44	**64.75**	41.98	62.18

Three observations follow from Table 18. (i) CAPE wins outright on five of nine cells and ties for first on one more, while CuPL wins one (SigLIP2/HandriseReadWrite, +5.6 pp), WaffleCLIP wins one (DFN/HandriseReadWrite, +1.7 pp), and PLAIN wins one (DFN/TeacherBehavior, +22.7 pp over CAPE). No prompt strategy is uniformly best across the nine $(\textit{backbone}, \textit{subset})$ cells, which is the same conclusion already drawn from the in-house Set A/B/C and the principle ablation. (ii) CuPL is systematically weakest on multi-label TeacherBehavior (36.6–72.4%): averaging ∼25 long descriptive sentences shifts the prototype toward generic "classroom photo" content and hurts the discriminative argmax used by Hit@1. (iii) The DFN/TeacherBehavior column where PLAIN (68.61%) far outperforms both CAPE (45.90%) and CuPL (42.65%) is consistent with the analyses in Sections 4.3.2 and 5.6: the DFN reverse-effect is a property of the DFN text encoder's poor alignment with action-prompt semantics on this subset and is not specific to CAPE-style hand-crafted prompts.

Overall, this comparison places CAPE on par with—and sometimes ahead of—two representative prompt-engineering baselines from the recent CLIP literature, and reinforces the central finding: prompt strategy selection dominates backbone selection, and no off-the-shelf strategy is universally safe in the SCB5 setting. Soft-prompt learning (e.g., CoOp [22]) requires few-shot supervision and is outside the zero-shot scope of this study.

5. Discussion

5.1. The Dominance of Prompt Design

The prompt ablation shows that the choice of prompt strategy affects performance more than the choice of model. For example, CLIP (OpenAI) with action prompts (70.25% Hit@1 on TeacherBehavior) outperforms SigLIP2 with label-only prompts (16.42%) by over 53 percentage points, despite SigLIP2 using a higher input resolution. This observation appears across all three benchmark sub-datasets and suggests that practical deployments should prioritize prompt engineering before considering model upgrades.

The non-monotonic relationship between prompt complexity and performance is scene-dependent (see Section 4.2.2 for per-model prompt profiles). One testable interpretation is that models trained on shorter, more diverse alt-text (e.g., OpenAI CLIP on WIT) may respond better to concise descriptions, while models trained on richer web text (SigLIP2 on WebLI) may benefit from more verbose prompts. The broader implication is that CLIP should be read as an analysis probe in this benchmark: absolute scores vary by prompt/model, but the subset-level difficulty pattern is comparatively stable.

5.2. Scene-Dependent Model–Prompt Interactions

A central finding enabled by the multi-scene benchmark is that **no single model–prompt combination is universally best**. SigLIP2+CAPE leads on TeacherBehavior (85.56%), LAION+action leads on HandriseReadWrite (84.56%), and DFN-CLIP+CAPE leads on BowTurnHead (93.27%). This diversity points to the conclusion that model selection in educational AI should be guided by the specific recognition scenario rather than by general-purpose benchmarks.

DFN-CLIP exhibits a strong model–task interaction: it ranks last on TeacherBehavior (45.86%) yet first on BowTurnHead (93.27%), both under CAPE prompts. DFN-CLIP's training pipeline aggressively filters web-crawled image–text pairs to retain only highly aligned examples [10]. One possible explanation is that this filtering biases its representation toward prototypical, visually unambiguous concepts. BowTurnHead's two classes (bow, turn-head) are precisely such concepts—spatially distinct postures with clear visual signatures—whereas TeacherBehavior's eight classes include frequent co-occurrences and subtle overlaps (e.g., "teacher + stand + screen" vs. "teacher + stand + blackboard") that require fine-grained disambiguation. This analysis suggests that data-filtering strategies during pre-training can have downstream consequences for domain-specific transfer, a dimension that remains underexamined in current VLM benchmarking.

EVA02-CLIP presents a different anomaly: despite reporting strong ImageNet accuracy through its masked-image-modeling pre-training objective [9], it achieves only 78.86% on TeacherBehavior (third) and 65.35% on BowTurnHead (last), whereas it leads on HandriseReadWrite (82.23%). As noted in Section 3.2 (†), EVA02-L/14 uses a MIM pre-training objective that is not architecturally equivalent to the other four contrastive backbones. The per-subset inconsistency, together with its exclusion from the paired bootstrap due to incomplete cache coverage, warrants caution in any cross-model interpretation of EVA02 results and motivates further investigation. ImageNet performance is dominated by object-centric categories, whereas classroom activity recognition requires understanding spatial relationships among people and objects—a mismatch that highlights the gap between generic visual representations and domain-specific behavioral recognition.

5.3. Cross-Family Validation Clarifies Robustness Across Model Families

These analyses add two points. First, MLLM comparisons confirm that the main claims do not depend on restricting analysis to CLIP-family encoders. Under the class-balanced weighted Macro-F1, both larger MLLMs in fact *outperform* the best CLIP reference (31.46% for `qwen3.5:27b` and 34.18% for `gemma4:26b` vs. 28.30%), even though the best CLIP reference still leads on weighted Hit@1; the size of this Macro-F1 advantage (about +3–6 pp) should however be read alongside the protocol asymmetry on TeacherBehavior noted in Section 3.6. Despite this gain, the MLLMs retain the central teacher-behavior problem: very high Hit@1 can still coexist with low class-balanced performance. Second, the EVA02 CAPE vs. best-prompt comparison shows that prompt-strategy disclosure is necessary, because the same backbone behaves very differently across prompt choices on different subsets. Together, these results strengthen the broader argument that benchmark conclusions in classroom zero-shot recognition should rely on multiple metrics and explicitly reproducible evaluation settings rather than isolated headline numbers.

5.4. Prompt Complexity Is Task-Dependent, Not Universally Optimal

The findings establish that **prompt effectiveness is task-dependent**. CAPE achieves the best Hit@1 on TeacherBehavior (85.56%, SigLIP2) and BowTurnHead (93.27%, DFN-CLIP), but *fails* on HandriseReadWrite, where simpler action prompts outperform CAPE by up to 15 pp (LAION: 84.56% action vs. 69.72% CAPE). This pattern is consistent across all five models on HandriseReadWrite: CAPE never wins.

This is attributed to the semantic structure of the target classes. HandriseReadWrite's three categories (hand-raise, read, write) are kinematically distinct—each maps to a single, well-defined body posture. CAPE's multi-sentence contextual descriptions, while helpful for disambiguating overlapping categories (e.g., "teacher + stand" vs. "teacher + blackboard"), introduce noise when categories are already well-separated. Conversely, TeacherBehavior's eight highly overlapping classes and BowTurnHead's subtle head-orientation distinction both benefit from CAPE's richer semantic grounding. In particular, TeacherBehavior is harder because many labels are semantically overlapping and context-coupled, so correct recognition depends less on isolated object cues and more on relational scene interpretation.

This finding yields a practical guideline: *richer prompt strategies are most valuable when classes are semantically overlapping or involve subtle distinctions*; for well-separated categories, simpler prompts are not only adequate but superior. The observed task dependency of prompt complexity informs the prompt-engineering literature for VLMs, where "more context is better" is often assumed without direct empirical checks.

The principle ablation in Table 8 further clarifies *why* CAPE helps in the cases where it does. Holding the prompt count fixed at $K = 3$, removing visual grounding (C1) or removing discriminative contrast (C3) each costs 10–18 pp Sample-F1 on TeacherBehavior and pushes BowTurnHead Hit@1 down toward the majority-class floor of 63.37% or, in the discriminative-contrast case, the degenerate floor of 36.63%. Removing only semantic diversity (C2) is much less harmful and even slightly helpful for one backbone. CAPE's gains over a one-line baseline therefore arise principally from class-specific, visually grounded discrimination rather than from averaging more prompts per class. This distinguishes CAPE from generic K-prompt ensembles that primarily rely on textual variety.

5.5. Robustness to Prompt Wording Is Also Task-Dependent

The robustness experiment adds a relevant point. Zero-shot prediction is deterministic for a fixed prompt set, but the *choice of wording* is itself a hidden degree of freedom. Set A vs. Set B differences therefore quantify a source of uncertainty in zero-shot reporting.

TeacherBehavior is again the hardest case: the gap between A(3) and B(3) is 11.27 pp for CLIP, 19.60 pp for LAION, 54.13 pp for SigLIP2, 25.12 pp for EVA02, and 23.68 pp for DFN-CLIP. The corresponding mixed-prompt variance is also highest on this sub-dataset, indicating that semantically overlapping multi-label scenes amplify prompt-phrasing sensitivity. By contrast, BowTurnHead is relatively stable once the action semantics are captured, especially for DFN-CLIP and EVA02-CLIP, whose mixed-prompt standard deviations remain below 4 pp.

HandriseReadWrite shows a different pattern: alternate Set B wording often outperforms the original CAPE prompts, even though CAPE itself does not beat the best action baseline in the main benchmark. This pattern suggests that for well-separated actions, robustness may come less from ensembling many semantically rich descriptions and more from selecting concise, visually direct phrasings. Practically, these results suggest that prompt-based zero-shot studies should report not only a best prompt score but also a perturbation-based robustness statistic such as Mix mean $\pm$ standard deviation, consistent with the emphasis on input-space robustness in LLM and AI agent security evaluation [11,12].

5.6. Multi-Label Hit@1 vs. Single-Label Metrics

The TeacherBehavior sub-dataset reveals a large evaluation gap: best Hit@1 reaches 85.56% while single-label macro F1 stays below 14.5%. This gap arises from the lenient multi-label criterion—any co-occurring label counts as correct—combined with the lossy primary-label reduction that creates severe class imbalance (Sections 3.5 and 4.5.2). The metric discrepancy is specific to multi-label scenarios and not an inherent limitation of zero-shot models.

The multi-label evaluation in Section 4.5.2 offers a more informative alternative: at $\tau = 0.20$, Sample-F1 reaches 64–66% and Macro-F1 reaches 55–60%, providing a more faithful assessment of class-discriminative ability. Importantly, EVA02-CLIP achieves the highest Macro-F1 (60.50%) despite ranking third in Hit@1—evidence that Hit@1 rankings can be misleading on multi-label data.

5.7. Supervised Baselines Contextualize Zero-Shot Value

The linear probing experiments (Section 4.4.2) provide context on evaluation metrics. On TeacherBehavior, zero-shot CAPE achieves higher Hit@1 than the supervised linear probe by up to 9.2 pp, which might suggest that labeled data is unnecessary. However, the multi-label comparison in Table 11 shows the opposite: supervised per-class binary classifiers achieve Sample-F1 of 88–90%, Macro-F1 of 68–78%, and Micro-F1 of 89–91%, far surpassing zero-shot CAPE's 60–66%, 49–61%, and 61–70% respectively. **The zero-shot Hit@1 advantage arises from lenient multi-label evaluation criteria**, where predicting any one of 3–5 co-occurring labels counts as correct. When evaluated on the ability to discriminate *all* eight activity classes, supervised learning with 8984 labeled images is superior.

On the single-label HandriseReadWrite dataset, the supervised advantage is consistent across all metrics (+15 pp accuracy), confirming that when clean training data is available and classes are well-separated, linear probing remains the more practical choice. The zero-shot approach retains value primarily in scenarios where labeled data is genuinely unavailable—the original motivation for zero-shot classroom recognition.

Practical implications of the supervised–zero-shot gap.

These results provide guidance on when zero-shot methods are practically justified. Given the substantial gap between supervised and zero-shot performance—LP Sample-F1 88–90% versus CAPE 60–66% on TeacherBehavior—zero-shot methods such as CAPE should not be viewed as a replacement for supervised learning when labeled data are available. Rather, CAPE is most valuable in low-supervision or deployment-constrained scenarios: it provides a deployment-ready baseline without any training overhead, requires no task-specific annotation, and can be applied immediately to new classroom settings. When annotation budgets allow, supervised linear probing on frozen features remains the more effective approach, as the linear probe experiments confirm across all five backbones.

5.8. Statistical Reliability

The bootstrap confidence intervals and paired tests (Section 4.4.3) reveal that several headline model differences lack statistical support. A key methodological point: overlapping marginal CIs do not imply that models perform equivalently, nor does a significant paired test automatically establish practical relevance. These distinctions caution against over-interpreting nominal gaps, and reinforce that the per-subset summary in Table 9 should be read alongside uncertainty estimates rather than as a hypothesis-tested global ranking.

5.9. Implications for Intelligent Classroom Systems

The findings carry concrete implications for practitioners deploying activity recognition in real educational environments. Zero-shot CLIP-family models require no annotated classroom data and can be deployed immediately on edge inference hardware [5] (e.g., campus SoC camera modules), making them attractive for resource-constrained settings. Critically, prompt strategies can be updated at inference time without retraining, which aligns well with the update constraints of low-power embedded systems where re-deploying a fine-tuned model is costly.

The present finding that richer prompts do not always improve performance provides direct guidance: for well-separated action classes (e.g., hand-raise vs. read vs. write), a concise action template suffices and avoids the overhead of assembling per-class prompt sets. Conversely, for semantically overlapping teacher-behavior categories, class-aware ensembling is worth the marginal text-encoding cost. Finally, the multi-label F1 and bootstrap methodology introduced here transfers directly to other intelligent classroom monitoring scenarios—such as engagement detection or group activity recognition—wherever single-label accuracy would mask class imbalance or co-occurrence structure. The security and privacy implications of deploying such systems in live educational environments merit further investigation [28].

5.10. Limitations

This study has several limitations. (1) All three sub-datasets originate from Chinese classroom contexts; generalization to other cultural or educational settings requires further investigation. (2) TeacherBehavior does not provide a canonical single-label target. The primary-label view used here is therefore only an auxiliary diagnostic, and stronger future benchmarks should prefer native multi-label evaluation throughout. (3) CAPE prompts are manually designed. The blind prompt evaluation (Section 4.5.1) and the literature-baseline comparison (Section 4.5.4, against CuPL [24] and WaffleCLIP [25] under an otherwise identical pipeline) provide independent validations and show that CAPE is on par with these generic prompt-engineering recipes; nevertheless, gradient-based soft-prompt methods such as CoOp [22] require few-shot supervision and were therefore outside this paper's zero-shot probe scope. For readability, exhaustive prompt and robustness tables are provided in the release package rather than in the main text. (4) The MLLM comparison uses a single deterministic prompt-and-parser protocol per image and therefore remains asymmetric with the CLIP-family evaluation on TeacherBehavior: MLLMs are forced to output one label, whereas CLIP-family models expose full similarity vectors that can also be thresholded in a multi-label setting. Future work should test prompt sensitivity, multi-turn prompting, calibrated answer parsing, and richer score extraction to determine how much of the observed gap is model-limited versus protocol-limited. (5) A single-frame evaluation protocol is used; real classroom behavior unfolds over time and would benefit from video-based models. In particular, several TeacherBehavior labels (e.g., "guide" vs. "answer" vs. "on-stage interaction") are intent-driven and difficult to separate from a single still frame, so part of the residual gap between the supervised LP ceiling ($\approx$90% Micro-F1) and a perfect classifier likely reflects genuine single-frame ambiguity rather than a fixable representational weakness. (6) The excluded Discuss sub-dataset (1 class) is trivially solved; richer group-interaction datasets with multiple sub-categories would provide more meaningful evaluation coverage. (7) The SCB sub-datasets are community Hugging Face releases that, to the best of the authors' knowledge, have not been accompanied by a peer-reviewed publication or formal technical report; researchers reusing this benchmark for publication are advised to verify the current provenance and citation guidance at the repository page. (8) All MLLMs in this study are evaluated as static single-turn classifiers under a constrained output protocol. Real-world deployment of vision–language models in intelligent classrooms could benefit from agentic capabilities such as multi-turn reasoning, tool use, and adaptive prompting; evaluating these capabilities requires protocols beyond the scope of this benchmark [5].

6. Conclusions

This benchmark study establishes two methodological lessons with broad applicability to VLM zero-shot evaluation in educational settings. First, top-1 accuracy on multi-label data is a lenient metric that systematically overstates model capability; class-discriminative alternatives and uncertainty estimates should accompany headline numbers in any benchmark report. Second, prompt complexity is not universally beneficial: richer class-aware prompts improve performance on semantically overlapping categories but degrade it on well-separated ones—a task dependency that is sign-consistent across all five evaluated backbones and refutes the intuition that more context is always better.

These lessons carry direct practical implications for the VLM benchmarking community. Any zero-shot classroom benchmark that reports only Hit@1 risks rewarding models that exploit label co-occurrence rather than genuine class discrimination. Paired bootstrap tests and multi-label F1 metrics together provide a more trustworthy evidence base for comparative claims, and both are straightforward to implement from released logit caches.

Future work should investigate calibrated multi-label prompting strategies, explore video-based temporal models capable of distinguishing intent-driven behaviors such as guiding versus answering, and extend the benchmark to richer multi-party group-interaction scenarios beyond the single-frame still-image protocol of the present study. Code and data are publicly available; see the Data Availability Statement.

Author Contributions: Conceptualization, Y.M. and L.Z.; methodology, Y.M. and L.Z.; software, L.Z.; validation, Y.M. and L.Z.; formal analysis, Y.M.; investigation, Y.M. and X.W.; data curation, Y.M., L.Z., and X.W.; visualization, L.Z.; writing—original draft preparation, Y.M.; writing—review and editing, Y.M. and L.Z.; supervision, L.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Education Department of Hunan Province, grant number HNJG-2022-0608.

Data Availability Statement: The three SCB sub-datasets used in this study (plus the excluded single-class Discuss set) are publicly available on Hugging Face at https://huggingface.co/datasets/wintonYF/SCB-Dataset (accessed on 27 April 2026). These datasets are externally released resources and are not owned by this study. The experiment code, prompt templates, cached image/text features (`data/feature_cache/`), the paired-bootstrap significance script (`scb5_zeroshot/paired_bootstrap.py`, source of Table 12), the three-principle ablation script (`scb5_zeroshot/cape_principle_ablation.py`, source of Table 8), the verbatim Set A and Set B prompts (`scb5_zeroshot/prompts/setAB_examples.json`), and the released summary result files are publicly available at https://github.com/zhanglizhuo/scb5-zeroshot (accessed on 27 April 2026).

Acknowledgments: The authors thank wintonYF and all contributors who created and maintained the public SCB dataset releases used in this study.

Conflicts of Interest: The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

References

1. Brophy, J. *Teaching*; International Academy of Education: Brussels, Belgium, 2000.

2. Anderson, L.W. *Increasing Teacher Effectiveness*, 2nd ed.; UNESCO International Institute for Educational Planning: Paris, France, 2004.
3. Dillenbourg, P.; Zufferey, G.; Alavi, H.; Jermann, P.; Do-Lenh, S.; Bonnard, Q.; Cuendet, S.; Kaplan, F. Classroom Orchestration: The Third Circle of Usability. In *Connecting Computer-Supported Collaborative Learning to Policy and Practice: CSCL2011 Conference Proceedings*; ISLS, 2011; pp. 510–517.
4. Liu, S.; Chen, Y.; Xie, H.; Wang, F.L. A Survey of Intelligent Classroom: Concepts, Technologies, and Applications. *Educ. Inf. Technol.* **2023**, *28*, 7127–7155.
5. Ma, W.; Guo, Y.; Han, Q.-L.; Zhou, W.; Zhu, X.; Xiong, J.; Wen, S.; Xiang, Y. Understanding Agentic AI: Algorithms and Infrastructure. *IEEE/CAA J. Autom. Sinica* **2026**, *13*, 776–795.
6. Radford, A.; Kim, J.W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; et al. Learning Transferable Visual Models From Natural Language Supervision. In Proceedings of the 38th International Conference on Machine Learning (ICML), Virtual, 18–24 July 2021; pp. 8748–8763.
7. Schuhmann, C.; Beaumont, R.; Vencu, R.; Gordon, C.; Wightman, R.; Cherti, M.; Coombes, T.; Katta, A.; Mullis, C.; Wortsman, M.; et al. LAION-5B: An Open Large-Scale Dataset for Training Next Generation Image–Text Models. *Adv. Neural Inf. Process. Syst.* **2022**, *35*, 25278–25294.
8. Tschannen, M.; Kumar, M.; Steiner, A.; Zhai, X.; Houlsby, N.; Beyer, L. SigLIP 2: Multilingual Vision–Language Encoders with Improved Semantic Understanding, Localization, and Dense Features. *arXiv* **2025**, arXiv:2502.14786.
9. Sun, Q.; Fang, Y.; Wu, L.; Wang, X.; Cao, Y. EVA-02: A Visual Representation for Neon Genesis. *arXiv* **2023**, arXiv:2303.11331.
10. Fang, A.; Jose, A.M.; Jain, A.; Schmidt, L.; Toshev, A.; Shankar, V. Data Filtering Networks. *arXiv* **2023**, arXiv:2309.17425.
11. Zhou, W.; Zhu, X.; Han, Q.-L.; Li, L.; Chen, X.; Wen, S.; Xiang, Y. The Security of Using Large Language Models: A Survey with Emphasis on ChatGPT. *IEEE/CAA J. Autom. Sinica* **2025**, *12*, 1–26.
12. Deng, Z.; Guo, Y.; Han, C.; Ma, W.; Xiong, J.; Wen, S.; Xiang, Y. AI Agents Under Threat: A Survey of Key Security Challenges and Future Pathways. *ACM Comput. Surv.* **2025**, *57*, 182.
13. Whitehill, J.; Serpell, Z.; Lin, Y.-C.; Foster, A.; Movellan, J.R. The Faces of Engagement: Automatic Recognition of Student Engagement from Facial Expressions. *IEEE Trans. Affect. Comput.* **2014**, *5*, 86–98. https://doi.org/10.1109/TAFFC.2014.2316163.
14. Donnelly, P.J.; Blanchard, N.; Olney, A.M.; Kelly, S.; Nystrand, M.; D'Mello, S.K. Words Matter: Automatic Detection of Teacher Questions in Live Classroom Discourse Using Linguistics, Acoustics, and Context. In Proceedings of the 7th International Learning Analytics & Knowledge Conference (LAK), Vancouver, BC, Canada, 13–17 March 2017; pp. 218–227. https://doi.org/10.1145/3027385.3027400.
15. Zhang, Z.; Li, Z.; Liu, H.; Cao, T.; Liu, S. Data-Driven Online Learning Engagement Detection via Facial Expression and Mouse Behavior Recognition Technology. *J. Educ. Comput. Res.* **2020**, *58*, 63–86. https://doi.org/10.1177/0735633119825575.
16. Li, X.; Chen, M.; Nie, F.; Wang, Q. A Multimodal Data Corpus for Audio-Visual Interest Detection in Classroom. *Front. Psychol.* **2022**, *13*, 903560. https://doi.org/10.3389/fpsyg.2022.903560.
17. Cherti, M.; Beaumont, R.; Wightman, R.; Wortsman, M.; Ilharco, G.; Gordon, C.; Schuhmann, C.; Schmidt, L.; Jitsev, J. Reproducible Scaling Laws for Contrastive Language–Image Learning. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Vancouver, BC, Canada, 18–22 June 2023; pp. 2818–2829. https://doi.org/10.1109/CVPR52729.2023.00276.
18. Zhai, X.; Mustafa, B.; Kolesnikov, A.; Beyer, L. Sigmoid Loss for Language–Image Pre-Training. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 2–6 October 2023; pp. 11975–11986. https://doi.org/10.1109/ICCV51070.2023.01100.
19. Wang, Z.; Wu, Z.; Agarwal, D.; Sun, J. MedCLIP: Contrastive Learning from Unpaired Medical Images and Text. In Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP), Abu Dhabi, United Arab Emirates, 7–11 December 2022; pp. 3876–3887. https://doi.org/10.18653/v1/2022.emnlp-main.256.
20. Liu, F.; Chen, D.; Guan, Z.; Zhou, X.; Zhu, J.; Ye, Q.; Fu, L.; Zhou, J. RemoteCLIP: A Vision Language Foundation Model for Remote Sensing. *IEEE Trans. Geosci. Remote Sens.* **2024**, *62*, 1–16. https://doi.org/10.1109/TGRS.2024.3390838.
21. Wang, M.; Xing, J.; Liu, Y. ActionCLIP: A New Paradigm for Video Action Recognition. *arXiv* **2021**, arXiv:2109.08472.
22. Zhou, K.; Yang, J.; Loy, C.C.; Liu, Z. Learning to Prompt for Vision–Language Models. *Int. J. Comput. Vis.* **2022**, *130*, 2337–2348. https://doi.org/10.1007/s11263-022-01653-1.
23. Zhou, K.; Yang, J.; Loy, C.C.; Liu, Z. Conditional Prompt Learning for Vision–Language Models. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, LA, USA, 18–24 June 2022; pp. 16816–16825. https://doi.org/10.1109/CVPR52688.2022.01631.
24. Pratt, S.; Covert, I.; Liu, R.; Farhadi, A. What Does a Platypus Look Like? Generating Customized Prompts for Zero-Shot Image Classification. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 2–6 October 2023; pp. 15645–15655. https://doi.org/10.1109/ICCV51070.2023.01438.

25. Roth, K.; Kim, J.; Koepke, A.S.; Vinyals, O.; Schmid, C.; Akata, Z. Waffling Around for Performance: Visual Classification with Random Words and Broad Concepts. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 2–6 October 2023; pp. 15700–15711. https://doi.org/10.1109/ICCV51070.2023.01443.
26. Deng, Z.; Ma, W.; Han, Q.-L.; Zhou, W.; Zhu, X.; Wen, S.; Xiang, Y. Exploring DeepSeek: A Survey on Advances, Applications, Challenges and Future Directions. *IEEE/CAA J. Autom. Sinica* **2025**, *12*, 872–893.
27. SCB-Dataset: Smart Classroom Behavior Dataset. Available online: https://huggingface.co/datasets/wintonYF/SCB-Dataset (accessed on 27 April 2026).
28. Deng, Z.; Sun, R.; Xue, M.; Ma, W.; Wen, S.; Nepal, S.; Xiang, Y. Hardening LLM Fine-Tuning: From Differentially Private Data Selection to Trustworthy Model Quantization. *IEEE Trans. Inf. Forensics Security* **2025**, *20*, 7211–7226.